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Advanced neural network architectures for tomato leaf disease diagnosis in precision agriculture

Hritwik Ghosh[1] · Irfan Sadiq Rahat[1] · Md. Mintajur Rahman Emon[2] · Md. Jisan Mashrafi[3] · Mohammed Abdul Al Arafat Tanzin[3] · Sachi Nandan Mohanty[1] · Shashi Kant[4]

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Abstract

This study addresses the critical challenge of detecting and classifying tomato leaf diseases using advanced deep learning technologies, a pivotal step in enhancing productivity within precision agriculture. We developed a novel diagnostic system capable of categorizing tomato leaves into ten distinct classes—nine corresponding to specific diseases and one for healthy leaves. The system incorporates a dataset of 6,000 images, which underwent extensive preprocessing, including resizing to 256 × 256 pixels, grayscale conversion, normalization, masking, and augmentation, to optimize input quality for the model. Our approach stands out through the integration of state-of-the-art neural network architectures—VGG19, Vision Transformer (ViT), EfficientNetV2, ConvNeXt—and a novel hybrid model specifically designed to leverage the strengths of diverse architectures. Performance evaluation demonstrated that the hybrid model outperformed all individual architectures, achieving an exceptional classification accuracy of 98%, ensuring robust disease detection under varying conditions. To enhance interpretability, Grad-CAM and LIME techniques were employed, highlighting critical image regions and influential features for classification. This comprehensive and innovative approach not only deepens understanding in plant pathology through automated systems but also sets a new benchmark for future advancements in plant disease detection leveraging machine learning.

Article Highlights

- Proposed a hybrid AI model combining Swin Transformer, Vision Transformer, and EfficientNetV2 for robust detection of tomato leaf diseases.
- Achieved exceptional classification accuracy, identifying healthy and diseased leaves across diverse conditions with nearly 98% precision.
- Offers a scalable solution for precision agriculture, enhancing early disease detection and reducing reliance on manual inspections, improving productivity.

Keywords Hybrid model · Grad-CAM · LIME techniques · Architectures · Vision Transformer

✉ Shashi Kant, skant317@gmail.com | [1]School of Computer Science and Engineering (SCOPE), VIT-AP University, Amaravati, Andhra Pradesh, India. [2]Department of Computer Science and Engineering, American International University-Bangladesh, Dhaka, Bangladesh. [3]Department of Computer Science and Engineering, Brac University, Dhaka, Bangladesh. [4]Department of Management, College of Business and Economics, Bule Hora University, Hagere Maryam, Ethiopia.

1 Introduction

The advent of precision agriculture has catalyzed the development of innovative technologies aimed at enhancing crop productivity and sustainability. Among the various challenges faced in this sector, detecting and classifying plant diseases stand out as critical tasks with significant implications for agricultural output and food security. Tomatoes, one of the most valuable horticultural crops globally, are particularly susceptible to a variety of leaf diseases that can dramatically affect their growth and yield.

Traditional methods of disease detection, which typically involve manual inspection and expert analysis, are not only labor-intensive but also prone to errors. As a result, there is an increasing demand for automated, accurate, and efficient disease detection systems. Leveraging the power of deep learning, our study introduces a robust diagnostic system designed to detect and classify tomato leaf diseases using advanced neural network architectures. This system represents a pivotal step toward automating disease detection in precision agriculture, thereby reducing reliance on human labor while increasing the accuracy and speed of diagnosis. Our approach utilizes a comprehensive dataset of 6,000 images representing both healthy and diseased tomato leaves, which underwent meticulous preprocessing to enhance model training efficacy.

Using Using several cutting-edge architectures—such as VGG19, Vision Transformer (ViT), EfficientNetV2, and ConvNeXt—this study introduces a novel hybrid model that demonstrates exceptional efficacy in detecting diseases across diverse scenarios. The significance of this research lies not only in its contribution to agricultural technology but also in its broader implications for food security and sustainable farming practices. By integrating advanced deep learning models with plant pathology, we aim to deliver a scalable solution adaptable to various crops and environmental conditions, paving the way for future advancements in agricultural technology.

The continuous evolution of machine learning models, particularly convolutional neural networks (CNNs) and their derivatives, has significantly enhanced the capability of diagnostic systems in detecting and classifying plant diseases—especially those affecting tomato leaves.

A groundbreaking study combined a CNN with a Convolutional Block Attention Module (CBAM) and Support Vector Machine (SVM), pioneering a real-time disease detection system suitable for mobile devices [1]. This model marked a substantial step toward deploying deep learning technologies in practical, field-ready applications. Another noteworthy contribution utilized Gray Level Co-occurrence Matrix (GLCM) techniques to extract detailed color and texture features, demonstrating how traditional image processing methods can bolster early disease detection and prevent significant agricultural losses [2]. This underlined the importance of integrating traditional image processing with modern machine learning frameworks to enhance diagnostic precision.

Further advancements showcased the potential of CNNs processed via Google Colab, highlighting their ability to analyze and extract intricate features, such as colors, textures, and edges, from images of tomato leaves [3]. The adaptability of CNNs was further exemplified by the customization of AlexNet architecture to meet computational constraints while maintaining high efficacy in feature extraction. This study optimized neural network architecture for practical applications, ensuring both power and efficiency [4].

The deployment of a modified AlexNet architecture on Android devices demonstrated the growing trend of making advanced diagnostic tools accessible and user-friendly for end-users, including farmers and agricultural technicians [5]. Meanwhile, the development of a DenseNet-77-based CornerNet model for accurate disease localization and classification illustrated the possibilities of combining sophisticated feature computation with robust detection frameworks [6]. This model exemplifies cutting-edge neural network applications in plant pathology, delivering high precision in disease detection.

In addition, the integration of the Inception Net model with advanced segmentation techniques, such as U-Net and Modified U-Net, highlighted a comprehensive approach to disease detection by effectively combining deep learning architectures with image segmentation to enhance overall diagnostic accuracy and performance [7].

Building upon these foundational studies, our research employs a hybrid model that integrates multiple advanced architectures, achieving unprecedented robustness in detecting and classifying tomato leaf diseases. This comprehensive approach leverages the computational power and adaptability of CNNs while enriching the model's capabilities through advanced image processing techniques, setting a new benchmark for automated plant disease diagnostic systems.

1.1 Background study

The agricultural sector is one of the pillars of global food security and economic stability. However, challenges such as plant diseases significantly threaten crop yield and quality, particularly in tomato farming. Tomatoes, being one of the most widely cultivated and consumed horticultural crops, are vulnerable to numerous leaf diseases that can drastically impact production. Traditional methods for detecting these diseases, which often involve manual inspection by experts, are labor-intensive, time-consuming, and prone to human error.

The advent of precision agriculture and advancements in deep learning have paved the way for innovative solutions to address these challenges. Automated disease detection systems powered by state-of-the-art neural network architectures offer a promising alternative. These systems can accurately and efficiently identify diseases, reducing reliance on human labor and enabling real-time decision-making in disease management.

This study explores advanced deep learning models such as Vision Transformer (ViT), EfficientNetV2, ConvNeXt, and hybrid architectures, leveraging their strengths to detect and classify tomato leaf diseases. By incorporating techniques like Grad-CAM and LIME, the research not only enhances model interpretability but also lays the groundwork for scalable and practical applications in precision agriculture. The findings aim to contribute significantly to sustainable farming practices, enhancing productivity while minimizing losses due to plant diseases.

1.2 Abbreviations and their full forms

To enhance clarity and comprehension, this section lists the abbreviations used throughout the paper along with their corresponding full forms. The table below serves as a quick reference for readers, ensuring consistency in understanding technical terms Table 1.

2 Related work

Zhao et al. developed a deep convolutional neural network integrated with an attention mechanism to diagnose tomato leaf diseases effectively. Their approach leverages residual blocks and attention modules to extract complex disease features, achieving an average identification accuracy of 96.81% on a tomato leaf dataset and 99.24% on a grape leaf dataset. The model demonstrated superiority in network complexity and real-time performance, outperforming existing methods in feature extraction and parameter efficiency [1].

Table 1 List of Abbreviations and Their Full Forms

Abbreviation	Full Form
AI	Artificial Intelligence
Grad-CAM	Gradient-weighted Class Activation Mapping
LIME	Local Interpretable Model-agnostic Explanations
CNN	Convolutional Neural Network
SVM	Support Vector Machine
GLCM	Gray Level Co-occurrence Matrix
YOLO	You Only Look Once
U-Net	Universal Net
MIADN	Multi-kernel Inception Aggregation Diffusion Network
BCNN	Bayesian Convolutional Neural Network
TL	Transfer Learning
MTDL	Multi-task Distillation Learning
CBAM	Convolutional Block Attention Module
ROC	Receiver Operating Characteristic
AUC	Area Under Curve
XAI	Explainable Artificial Intelligence
mAP	Mean Average Precision
F1-Score	Harmonic Mean of Precision and Recall

Hossain et al. explored transformer-based approaches for tomato leaf disease classification, comparing four models: MaxViT, EANet, CCT, and PVT. Their findings revealed that MaxViT outperformed others with a 97% accuracy on a multiclass dataset, surpassing EANet (89%), CCT (91%), and PVT (93%). The study highlighted MaxViT's smoother learning curve and efficacy in aggregating multi-scale attention, making it a robust solution for disease diagnosis given sufficient computational resources [2].

Kumar et al. utilized YOLOv5 and YOLOv7 models with transfer learning for tomato leaf disease detection. By incorporating a YOLO-YOLO cascaded ensemble, their approach enhanced multiscale feature detection, achieving a mean average precision (mAP) of 98.8% at a 0.5 threshold and 98.7% at 0.95. The ensemble model demonstrated high precision (95.8%) and recall (96.8%), proving its efficacy in accurately identifying and localizing tomato leaf diseases under natural conditions [3].

Babu et al. introduced a method combining the LeNet-5 CNN model and Otsu multi-thresholding with an optimization algorithm for tomato leaf disease diagnosis. Using the Plant Village dataset, their approach outperformed Xception, ResNet50, and VGG16 in classification accuracy. Real-time testing with unseen diseased images demonstrated the model's strong generalization ability, showcasing its practical applicability for precise disease identification [4].

A recent study proposed a deep learning-assisted SVM approach for tomato leaf disease classification, utilizing features extracted from GoogleNet, AlexNet, and ResNet-50. The model achieved 96.99% accuracy on the PlantVillage dataset, showcasing the effectiveness of combined deep features for precision agriculture. Evaluated using accuracy, precision, sensitivity, and F-scores, this method highlights its potential to transform disease detection by providing accurate and efficient solutions [5].

Seno et al. introduced Bayesian convolutional neural networks (BCNNs) for tomato leaf disease diagnosis to address overfitting in small datasets and quantify prediction uncertainty. Their BCNN model demonstrated improved robustness compared to conventional CNNs and provided confidence estimates for predictions. This approach enhances the reliability and practicality of plant disease diagnostic models, especially in scenarios with limited data availability [6].

Perveen et al. proposed a multiscale U-network model for concurrent segmentation and diagnosis of tomato leaf lesions. By integrating multi-scale residual modules and a Classifier and Bridge (CB) module, their approach achieved an average pixel accuracy of 99.2% and demonstrated robustness against brightness changes and noise. The model's computational efficiency and resilience highlight its potential for practical applications in agricultural disease management [7].

Gadade and Kirange proposed an automated system for tomato leaf disease detection and severity measurement using a segmentation-based approach. Their method integrates preprocessing, segmentation, feature extraction (color, texture, and shape), and classification to effectively analyze and classify infected regions. The system's performance demonstrates its potential to assist farmers in accurately detecting and measuring disease severity, promoting its use in agricultural practices [8].

Drake et al. integrated explainable computer vision techniques with deep transfer learning for tomato leaf disease diagnosis, using five pre-trained models: Vision Transformers, EfficientNetBO, ResNet50, VGG16, and VGG19. Their approach achieved high accuracies, with Vision Transformers leading at 98.5%. To enhance model transparency, they applied Explainable Artificial Intelligence (XAI) techniques, such as LRP, Grad-CAM, and SHAP, making the models' decision-making process more interpretable and reliable for practical use by farmers [9].

Paramesh and Narayana proposed an improved Convolutional Neural Network (CNN) model for tomato leaf disease diagnosis, achieving a high accuracy of 97% in detecting diseases. Their study focused on a mobile application, AgroDeep, which collects and uploads images of infected plant leaves, including those of tomatoes, grapes, soybeans, sugarcane, cotton, and onions. The research demonstrated how CNN-based techniques can enhance the detection and classification of plant diseases, contributing to food production in agriculture [10].

Karthika et al. proposed a deep learning approach using Convolutional Neural Networks (CNNs) for automated tomato plant leaf disease diagnosis. The study highlighted the challenges faced in plant disease detection, such as dataset collection, inter-class similarity, and real-time detection. By training a CNN model on a dataset of healthy and diseased tomato leaves, the researchers developed a system capable of accurately classifying new images as healthy or infected. This technique provides a fast and effective solution for disease identification in tomato crops, contributing to precision agriculture [11].

Gharghory proposed an automatic tomato leaf disease detection system based on fine-tuning and transfer learning of pre-trained deep Convolutional Neural Networks (CNNs), including AlexNet, VGG-16 Net, and SqueezeNet. The study compared the performance of these networks on two datasets: a small one with four diseases and a large one with nine diseases and healthy leaves. The experimental results showed that AlexNet outperformed other networks, achieving a classification accuracy of 97.4% with lower training time. VGG-16 Net, although providing the highest accuracy, consumed

more time during training. The findings highlight the potential of using transfer learning with pre-trained networks for efficient and accurate tomato leaf disease diagnosis [12].

Sun et al. introduced a Multi-kernel Inception Aggregation Diffusion Network (MIADN) for real-time tomato leaf disease detection, addressing the challenge of scale variations in diseased tomato leaves. The proposed model features a Multi-kernel Inception Module (MKIM), designed to extract multi-scale object features, enhancing the model's ability to fuse and represent features effectively. The authors also incorporated the FasterNet network at the feature extraction stage to preserve feature diversity and improve target feature extraction. Experimental results demonstrated that the MIADN model achieved a mean average precision (mAP50) of 96.6%, surpassing both the baseline and advanced YOLOv9s models, providing a highly effective solution for tomato disease detection and high-quality tomato cultivation [13].

Hou et al. proposed an improved deep convolutional neural network (CNN) for tomato leaf disease diagnosis, designed specifically for real-life agricultural environments with complex backgrounds and diverse disease conditions. The network integrates residual blocks, a multiscale feature extraction module for capturing lesions at various scales, and an attention module to enhance the extraction of lesion features by capturing cross-dimensional interactions. Experimental results demonstrated an average identification accuracy of 88.64% on a dataset with a complex backdrop. The proposed method outperformed other models in terms of network complexity and real-time performance. Additionally, the attention module contributed to more accurate disease spot recognition with fewer parameters compared to state-of-the-art approaches, making the model a high-performance solution for crop disease diagnosis in real agricultural settings [14].

Liu et al. introduced a novel multi-task distillation learning (MTDL) framework for tomato leaf disease recognition, addressing challenges in symptom variations, limited labeled data, and model complexity. Their approach incorporates knowledge disentanglement, mutual learning, and knowledge integration through a multi-stage strategy, enabling the framework to simultaneously tackle classification and severity prediction tasks. The MTDL-optimized EfficientNet outperformed the single-task ResNet101 by achieving higher classification accuracy (improved by 0.68%) and severity estimation (improved by 1.52%) while using only 9.46% of the parameters. This framework highlights its potential for intelligent agriculture applications, offering both improved performance and reduced model complexity [15].

Bouni et al. introduced a Computer-Aided Diagnosis system for automated detection and classification of tomato leaf diseases, integrating traditional handcrafted features with deep learning techniques. Their system involves preprocessing, feature extraction, feature fusion, and classification stages, utilizing enhancement filters and segmentation algorithms to isolate regions of interest (ROI) in tomato leaf images. The features, based on the ABCD rule (Asymmetry, Borders, Colors, and Diameter), are combined with outputs from a Convolutional Neural Network (CNN) pretrained on ImageNet. To address data imbalance, the authors propose a novel evaluation method, improving classification accuracy by 15%, achieving an overall accuracy of 92% in field tests. Their approach sets a new standard for precision in agricultural diagnostics by synergizing classical feature engineering with modern machine learning techniques [16].

Vini and Rathika proposed TrioConvTomatoNet, a novel deep convolutional neural network (CNN) architecture designed for fast and accurate tomato leaf disease classification. The architecture includes a 3-series convolution layer at each stage of the feature extraction process to efficiently capture and integrate information from leaf images. Through extensive experiments, the method achieved an accuracy of 99.39% on both existing and real-time datasets, outperforming state-of-the-art methods in terms of both accuracy and speed, making it suitable for practical agricultural applications [17].

Srivastava et al. proposed a hybrid approach for tomato leaf disease detection using MobileNetV2 for feature extraction and Extreme Learning Machine (ELM) for classification. The TLMV2-ELM model achieved impressive results, with an accuracy of 99% and a loss of 0.06, outperforming existing methods in disease detection. This approach leverages the efficiency of transfer learning and the fast training capabilities of ELM for enhanced classification performance [18].

Sivalingam and Varma (2007) developed a set of common primers and four virus-specific primers for the detection of begomoviruses associated with tomato leaf curl disease (TomLCD) in India. Their study revealed that out of 26 TomLCD samples tested, 22 were positive with common primers, and 16 were positive for specific begomoviruses (ToLCNDV, ToLCBV, ToLCKV, and ToLCGV). The research highlighted the widespread distribution of ToLCNDV and ToLCBV in northern and southern India, while ToLCKV and ToLCGV had a broader presence. Additionally, mixed infections were observed in some samples, and some plants exhibited typical symptoms but tested negative, suggesting the presence of uncharacterized viruses [19].

Indu and Priyadharsini [20] proposed an automatic tomato leaf disease identification system using a pre-trained AlexNet architecture optimized by the Crossover-based Wind-driven Optimization (CROWDO) algorithm. This approach enhances classification accuracy, achieving 99.86% for identifying diseases like Leaf Mold, Early Blight, and Tomato Yellow

Leaf Curl. The CROWDO-optimized model also reduces overfitting, training time, and energy consumption, outperforming other pre-trained models such as VGG16 and GoogleNet. The system enables real-time identification of disease areas, assisting farmers in early detection and management of pest infestations [20].

Cheemaladinne and K. [21] introduced a novel method for tomato leaf disease detection using the "VARMAx-CNN-GAN Integration" framework, which combines convolutional neural networks (CNNs) and generative adversarial networks (GANs) with vector autoregressive moving average processes (VARMAx). This integrated approach offers proactive disease detection by classifying diseases early, generating synthetic images to expand the dataset, and enhancing the model's generalization. The model improves accuracy, precision, recall, and AUC, offering better disease localization and severity estimation. It also shows potential for disease prevention, advancing both diagnosis and management in agriculture [21].

Paul et al. [22] proposed a lightweight custom convolutional neural network (CNN) model with transfer learning (TL) from VGG-16 and VGG-19 for accurate tomato leaf disease classification. The model was trained using eleven disease classes, including a healthy class, and applied data augmentation techniques to enhance performance. The ablation study helped optimize parameters, achieving a high accuracy and recall of 95.00%. The best-performing model was incorporated into a real-time web and Android-based end-to-end system for tomato cultivators, providing a practical tool for disease diagnosis and management [22].

Khan et al. [23] introduced seven Bayesian optimized deep hybrid learning models for automated classification of tomato leaf diseases, leveraging the combination of deep learning and machine learning techniques. The study utilized a customized Convolutional Neural Network (CNN) for feature extraction and various classical machine learning algorithms (RF, XGBoost, GNB, SVM, MLR, KNN) with stacking for classification. The Boruta feature filtering layer was incorporated to capture significant features. The CNN-Stacking model achieved impressive performance, with precision, recall, F1-score, MCC, and accuracy values of 98.53%, 98.53%, 98.53%, 98.53%, and 98.27%, respectively. The model also demonstrated high efficiency, requiring only 0.174 s for testing. This approach is computationally efficient and robust, offering real-time disease diagnosis potential for farmers through a smartphone app [23].

Liu and Wang [24] proposed a MobileNetv2-YOLOv3 model for the early recognition of tomato gray leaf spot disease, balancing accuracy and real-time detection. By introducing the GIoU loss function and utilizing transfer learning with mixup training, the model achieved high recognition accuracy, with F1 scores of 94.13% and 93.24% on specific test sets. The model demonstrated fast detection speeds, processing up to 246 frames/s on GPU, making it suitable for mobile app deployment to support farmers in real-time disease diagnosis [24].

S and H [25] investigated tomato leaf disease detection using machine learning techniques such as Fuzzy Support Vector Machine (Fuzzy-SVM), Convolutional Neural Network (CNN), and Region-based Convolutional Neural Network (R-CNN). They utilized various image processing methods, including image scaling, color thresholding, and segmentation, along with features like gradient local ternary pattern and Zernike moments for training. The R-CNN classifier outperformed the others, achieving an accuracy of 96.735%, making it the most effective model for tomato leaf disease prediction [25].

Lee et al. [26] proposed an improved YOLOv5m model for tomato leaf disease recognition, integrating C3NN modules and Bidirectional Feature Pyramid Network (BiFPN) architecture. The C3NN modules combined several soft attention mechanisms, such as CBAM, SE, ECA, and CA, to enhance feature representation and detection accuracy. The model achieved a precision of 87.764%, recall of 87.201%, F1 score of 87.482%, and mAP.5 of 90.401%, outperforming both YOLOv5m and Faster-RCNN models. These improvements highlight the model's potential in advancing plant disease detection technology [26].

Rajan et al. [27] developed a mobile application for tomato leaf disease identification using a Convolutional Neural Network (CNN) model. This approach helps farmers upload images of tomato leaves to identify the type of disease and receive treatment suggestions. The model, trained on a tomato leaf image dataset, aims to address the challenges of diagnosing plant diseases with similar symptoms. The results of this study offer valuable insights for plant science, promoting the development of similar models to assist farmers in disease detection and management [27].

Romero et al. [28] studied the effects of *Clavibacter michiganensis* subsp. *michiganensis*, the pathogen causing bacterial wilt and canker in tomato plants, on plant growth and water parameters. Inoculated tomato plants exhibited reduced growth, smaller leaf area, and lower biomass compared to controls. The pathogen decreased stem hydraulic conductivity, leading to wilting of the leaves. Early detection of the disease was possible by using the NDWI index, which showed spectral reflectance changes indicative of water content alterations before visible symptoms appeared. This index offers a potential tool for early, non-destructive diagnosis of bacterial cancer [28].

Kaur and Shalu [29] proposed a machine learning-based approach using the InceptionV3 model to detect tomato leaf diseases at an early stage, helping prevent further plant damage. Their method achieved a classification accuracy of 81%. The study also involved a comparative analysis of existing models to identify the most effective one for diagnosing

Table 2 Summary of the Related work

Study	Methodology	Dataset	Accuracy	Key Findings
Zhao et al. [1]	Deep CNN with Attention Mechanism	Tomato and Grape Leaf Datasets	96.81% (Tomato), 99.24% (Grape)	Residual blocks and attention modules for feature extraction, superior network complexity and performance
Hossain et al. [2]	Transformer-based Models (MaxViT, EANet, CCT, PVT)	Multiclass Tomato Leaf Dataset	97% (MaxViT)	MaxViT outperformed other models, smoother learning curve, and better multi-scale attention aggregation
Kumar et al. [3]	YOLOv5 & YOLOv7 with Transfer Learning	Tomato Leaf Dataset	mAP: 98.8% (0.5 threshold), 98.7% (0.95 threshold)	YOLO-YOLO cascaded ensemble for improved multiscale feature detection
Ramesh Babu et al. [4]	LeNet-5 CNN + Otsu Multi-thresholding	Plant Village Dataset	N/A	Outperformed Xception, ResNet50, and VGG16 in classification accuracy, strong generalization ability
Recent Study [5]	SVM with Deep Features from GoogleNet, AlexNet, and ResNet-50	PlantVillage Dataset	96.99%	High accuracy using combined deep features for precision agriculture
Seno et al. [6]	Bayesian CNN (BCNN)	Tomato Leaf Dataset	N/A	Improved robustness compared to conventional CNNs, addressed overfitting in small datasets
Perveen et al. [7]	Multiscale U-network	Tomato Leaf Dataset	99.2% (Pixel Accuracy)	Concurrent segmentation and disease diagnosis, robust against brightness changes and noise
Gadade and Kirange [8]	Segmentation-based Approach	Tomato Leaf Dataset	N/A	Classified infected regions and measured disease severity, accurate disease severity detection
Drake et al. [9]	Explainable Deep Transfer Learning	Tomato Leaf Dataset	98.5%	Vision Transformers and XAI techniques for model transparency, high accuracy
Paramesh and Narayana [10]	Improved CNN	Tomato, Soybean, Cotton, etc	97%	Mobile app AgroDeep for disease classification, improves food production
Karthika et al. [11]	CNN-based Approach	Tomato Leaf Dataset	N/A	Accurate classification for disease detection, fast and effective solution for real-time detection
Gharghory [12]	Transfer Learning with Pretrained CNNs (AlexNet, VGG-16, SqueezeNet)	Tomato Leaf Dataset	97.4% (AlexNet)	AlexNet outperformed others in classification accuracy with lower training time
Sun et al. [13]	Multi-kernel Inception Aggregation Diffusion Network (MIADN)	Tomato Leaf Dataset	96.6% (mAP50)	Real-time disease detection, enhanced feature extraction with multi-scale object features
Hou et al. [14]	Improved CNN with Attention	Tomato Leaf Dataset	88.64%	Network complexity optimized for real-world agricultural environments
Liu et al. [15]	Multi-task Distillation Learning (MTDL)	Tomato Leaf Dataset	Improved by 0.68% in classification accuracy	Simultaneous classification and severity prediction, optimized EfficientNet model

Table 2 (continued)

Study	Methodology	Dataset	Accuracy	Key Findings
Bouni et al. [16]	Computer-Aided Diagnosis System	Tomato Leaf Dataset	92%	Synergized classical feature engineering with deep learning, improved field test accuracy
Vini and Rathika [17]	TrioConvTomatoNet	Tomato Leaf Dataset	99.39%	Fast and accurate classification, outperformed existing methods in accuracy and speed
Srivastava et al. [18]	MobileNetV2 + Extreme Learning Machine	Tomato Leaf Dataset	99%	High accuracy and low training time
Sivalingam and Varma [19]	Primers for Begomovirus Detection	Tomato Leaf Samples	N/A	Focused on Tomato Leaf Curl Disease, detecting mixed virus infections
Indu and Priyadharsini [20]	CROWDO-Optimized AlexNet	Tomato Leaf Dataset	99.86%	Optimized CROWDO algorithm to enhance accuracy and reduce overfitting
Cheemaladinne and K. [21]	VARMAx-CNN-GAN Integration	Tomato Leaf Dataset	Improved accuracy, precision, recall, AUC	Synthetic image generation to enhance dataset and improve disease detection
Paul et al. [22]	Custom CNN with Transfer Learning	Tomato Leaf Dataset	95%	Real-time mobile app for disease detection and management
Khan et al. [23]	Bayesian Optimized Deep Hybrid Learning	Tomato Leaf Dataset	98.27%	Efficient, real-time disease detection using CNN-Stacking model
Liu and Wang [24]	MobileNetv2-YOLOv3	Tomato Gray Leaf Spot Dataset	94.13%	High recognition accuracy, fast detection for mobile app deployment
S and H [25]	Fuzzy SVM, CNN, R-CNN	Tomato Leaf Dataset	96.735%	R-CNN outperformed other models, best performance for disease prediction
Lee et al. [26]	Improved YOLOv5m with C3NN and BiFPN	Tomato Leaf Dataset	90.40% (mAP)	Enhanced feature representation, improved detection accuracy
Rajan et al. [27]	CNN-based Mobile App	Tomato Leaf Dataset	N/A	Real-time disease detection for farmers via a mobile app
Romero et al. [28]	NDWI Index	Tomato Plant Dataset	N/A	Early detection of bacterial wilt and canker through spectral reflectance changes
Kaur and Shalu [29]	InceptionV3 Model	Tomato Leaf Dataset	81%	Early-stage detection of tomato leaf diseases, proposed method showed high classification accuracy
Kokate et al. [30]	CNN Model	Tomato Leaf Dataset	95.53%	Classified 14,240 images of nine disease classes, crucial for early disease detection

leaf disorders in tomatoes. The approach aims to improve plant protection by enabling timely detection and treatment, which is crucial for ensuring crop productivity and minimizing economic losses [29].

Kokate et al. proposed a convolutional neural network (CNN) model for diagnosing tomato leaf diseases, utilizing 14,240 images representing nine distinct disease classes. The model achieved an average classification accuracy of 95.53%. This automated approach helps in the early (Table 2) identification of diseases, crucial for controlling crop damage and maximizing yield potential [30].

Paithane and Wagh [31] introduced a novel modified kernel fuzzy c-means (MKFCM) algorithm for cotton leaf spot detection. Traditional clustering techniques often struggle with noise and variations in image data, which this modified kernel-based approach aims to mitigate. The authors reported improved segmentation accuracy compared to conventional fuzzy c-means clustering, making it more effective for precise agricultural disease detection.

In the field of medical imaging, Paithane and Kakarwal [32] proposed LMNS-Net, a Lightweight Multiscale Novel Semantic-Net deep learning approach for automatic pancreas image segmentation in CT scans. The LMNS-Net incorporates multiscale feature extraction to enhance segmentation performance while maintaining computational efficiency. The study demonstrated superior results compared to standard deep learning architectures, highlighting the potential of lightweight models in medical image processing.

Further advancing pancreas segmentation techniques, Paithane [33] developed an optimized multiscale feature hybrid-net deep learning approach. This model integrates multiscale feature extraction with hybrid deep learning strategies to improve segmentation precision. The study emphasized the importance of balancing computational efficiency with high segmentation accuracy, a crucial factor in real-world clinical applications.

Additionally, Kakarwal and Paithane [34] investigated automatic pancreas segmentation using a ResNet-18 deep learning approach. The study leveraged ResNet-18's residual connections to enhance feature propagation and gradient flow, resulting in improved segmentation outcomes. Their findings underscored the effectiveness of deep residual networks in medical image analysis, especially for complex organ segmentation tasks.

3 Proposed method

3.1 Dataset description

The dataset utilized in this study comprises 6,000 high-quality images in PNG format, meticulously curated to represent a diverse range of tomato leaf conditions. These images are evenly distributed across ten distinct classes, with each class corresponding to a specific disease type or a healthy state. The classes include nine different tomato leaf diseases, as well as one class representing healthy, disease-free leaves. A ground truth image corresponds to each image in the dataset; this image is used to accurately manually segment the parts of the leaf that are healthy and those that show symptoms of sickness. This pairing is critical for training supervised models, as it provides an exact benchmark for the models to learn from and against which their predictions are evaluated.To prepare the dataset for effective machine learning model training, several preprocessing steps were implemented. To make sure the input size was consistent, all the photos were first downsized to 256×256 pixels. The images then underwent grayscale conversion to reduce computational demands and focus the model's learning on textural and structural features rather than color. Normalisation was applied to scale pixel values to a range between 0 and 1, enhancing the numerical stability and performance of the learning algorithms. Additionally, image masking techniques were employed to isolate the region of interest by reducing background noise, thereby allowing models to concentrate on relevant features. Finally, to augment the dataset and improve the model's generalization capabilities, various image augmentation techniques such as rotations, scaling, and horizontal flipping were applied. This comprehensive dataset setup not only facilitated the training of deep learning models, but also ensured that the models were exposed to a wide variety of scenarios, mimicking real-world conditions under which tomato plants might manifest disease symptoms or maintain health.

3.2 Image preprocessing

Effective image preprocessing is crucial for enhancing the performance of DL models, particularly in tasks involving high variability such as plant disease detection from images. In this study, several preprocessing techniques were systematically applied to prepare the dataset of tomato leaf images before feeding them into the convolutional

neural networks.The primary goals were to standardize the images, reduce computational load, and augment the data to prevent overfitting and improve model generalization.

- *Image Resizing* All images were resized to a uniform resolution to ensure consistency across the dataset and reduce computational demands during training. Specifically, each image was resized to 256 × 256 pixels using the following interpolation technique:

$$I\prime(x\prime, y\prime)I(\frac{x\prime \times W}{W\prime}, \frac{y\prime \times H}{H\prime}) \tag{1}$$

In Eq. (1), I(x′, y′) represents the pixel intensity value at the coordinates (x′, y′) in the resized image, while I (x, y) denotes the pixel intensity value at the coordinates (x, y) in the original image. The variables W and H are the original width and height of the image, respectively, while W′ and H′ are the desired width and height, set to 256 pixels each for uniform resizing in this study. The interpolation function f computes the intensity value for fractional pixel locations, with common methods like bilinear or bicubic interpolation. The coordinates x and y in the original image are scaled and mapped to the transformed coordinates x′ and y′ in the resized image. This ensures consistent dimensions across all images in the dataset, reducing computational complexity during training.

- *Grayscale Conversion* To focus the learning on textural and structural features rather than color, which is less relevant for identifying disease patterns, images were converted to grayscale. This reduces the data dimensionality from three color channels to a single intensity channel, simplifying the model's complexity:

$$\boldsymbol{G}(\boldsymbol{x}, \boldsymbol{y}) = 0.298 \cdot \boldsymbol{R}(\boldsymbol{x}, \boldsymbol{y}) + 0.587 \cdot \boldsymbol{G}(\boldsymbol{x}, \boldsymbol{y}) + 0.114 \cdot \boldsymbol{B}(\boldsymbol{x}, \boldsymbol{y}) \tag{2}$$

where G(x,y) denotes the grayscale value at pixel (x,y), and R(x,y), G(x,y), and B(x,y) are the red, green, and blue channel intensities, respectively. This formula reflects the human eye's differing sensitivity to these colors, emphasizing more significant components for perception.

- *Image Normalization* Normalizing the pixel values of the images to a range of [0, 1] enhances the numerical stability of the neural network by maintaining a consistent scale:

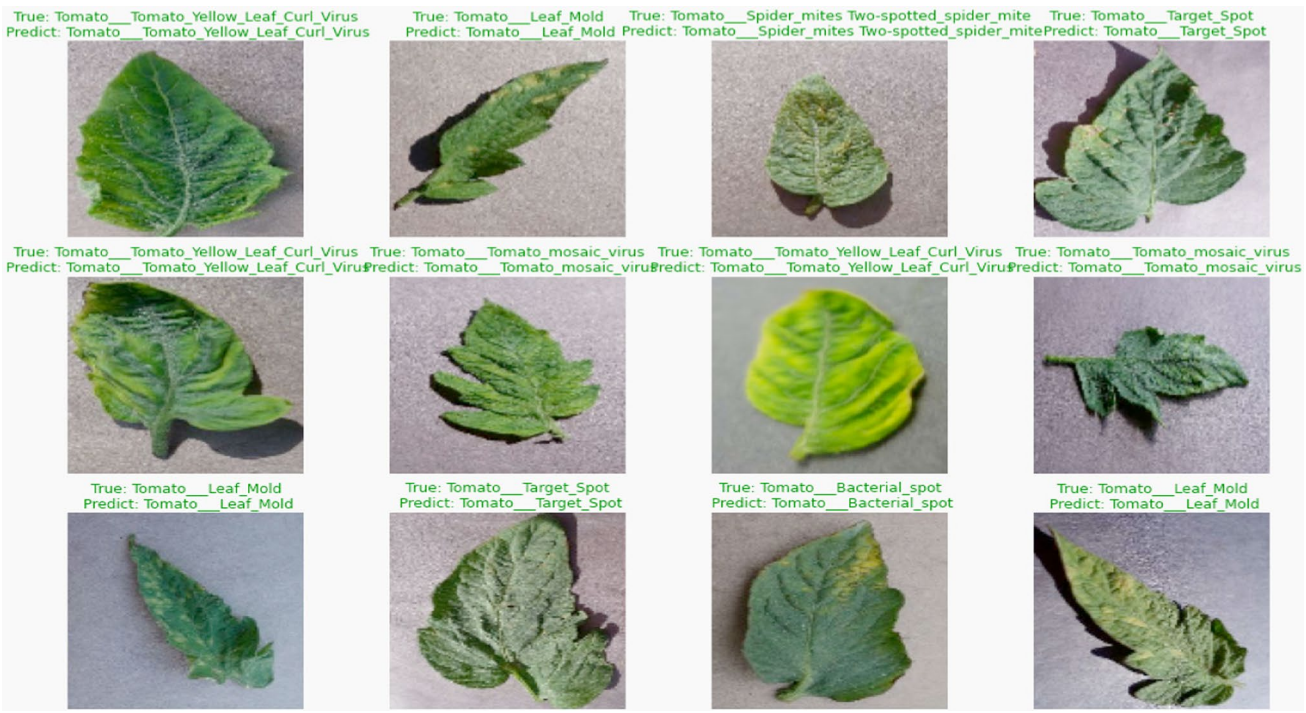


Fig. 1 Sample Image of Tomato Leaf

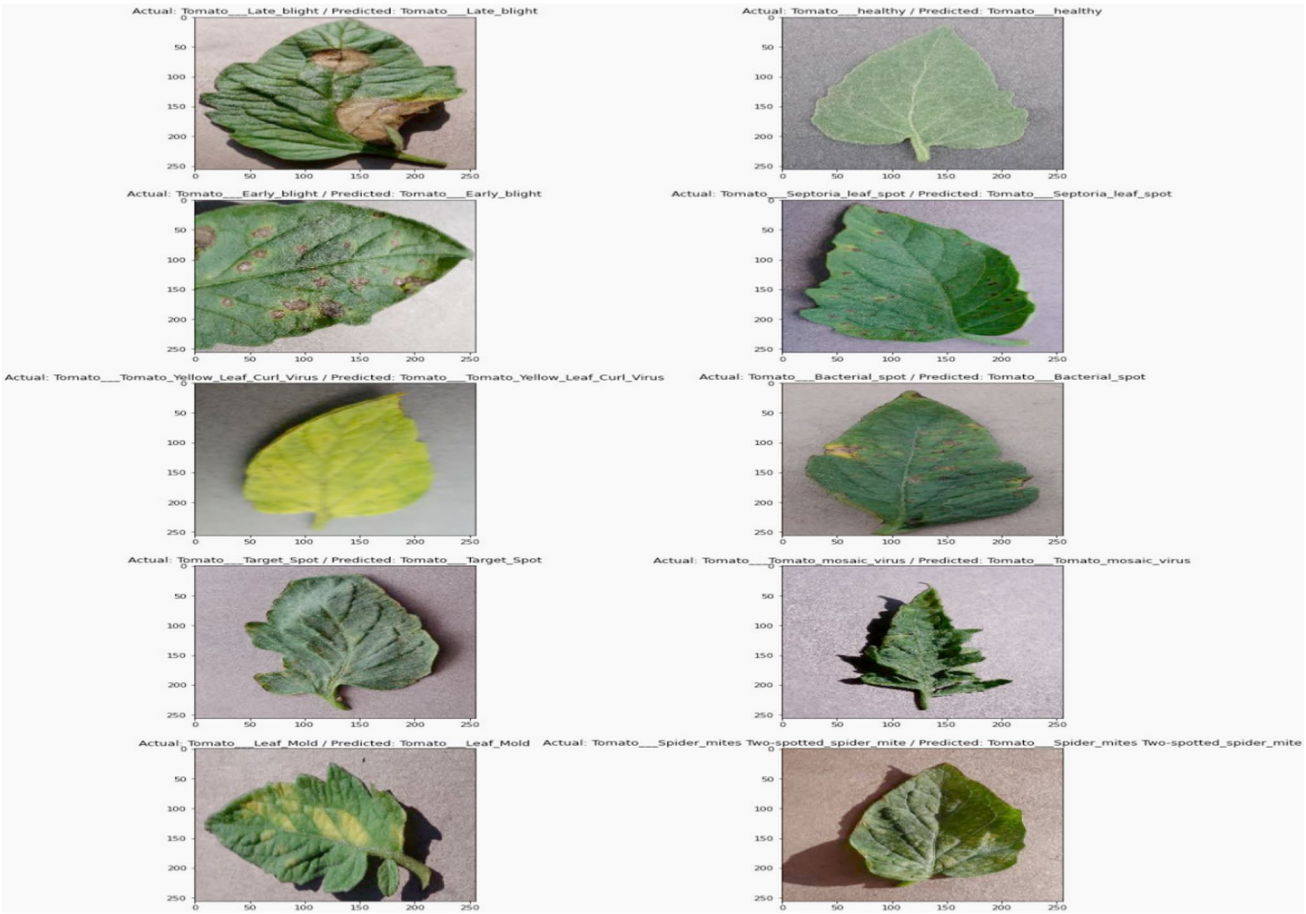


Fig. 2 Sample Tomato Leaf Images with disease conditions

Table 3 Parameter of Data Augmentation

Parameter	Suggested Value
Rescale	1/255
Rotation Range	30
Width Shift Range	0.1
Height Shift Range	0.1
Horizontal Flip	True
Vertical Flip	False
Zoom Range	[0.8, 1.2]

$$I_{norm} = \frac{I_{gray} - min(I_{gray})}{max(I_{gray}) - min(I_{gray})} \tag{3}$$

This operation ensures that all input features (pixel values) contribute equally to the analysis, preventing any single feature from dominating due to its scale.

- *Image Masking* To reduce background noise and focus the model on the regions of interest (i.e., the leaf areas), image masking techniques were applied. This involved creating binary masks that isolate the foreground (leaf) from the background based on color and intensity thresholds.

$$\mathbf{M(x,y)} = \left\{ \begin{array}{c} \mathbf{I(x,y)} \\ 0 \end{array} \right\} \begin{array}{c} \mathbf{if\, I(x,y)} \varepsilon \boldsymbol{ROI} \\ \mathbf{otherwise} \end{array} \tag{4}$$

4 Explanation

- ❖ M(x,y) represents the masked image, where only the pixels within the **Region of Interest (ROI)** are preserved.
- ❖ I(x,y) denotes the intensity or pixel value of the original image at coordinates (x,y).
- ❖ The condition (x,y) ∈ ROI indicates that the pixel at position (x,y) belongs to the predefined **Region of Interest**.
- ❖ If the pixel is within the ROI, its original intensity I(x,y) is retained.
- ❖ If the pixel is outside the ROI, it is assigned a value of **0**, effectively masking that region.

For this study, the intensity threshold T is set to [50,200], ensuring **(**Figs. 1, 2**)**that pixels within this range are included in the mask.

4.1 Data augmentation

Data augmentation is an effective method for expanding the variety of your training set through the use of arbitrary, realistic image modifications. This method enhances the model's generalizability to previously unseen data and helps keep it from over-fitting.Here, we detail the parameters used for data augmentation along with their effects on the images:

Re-scale is used to normalize the pixel values of the images. Normally, pixel values range from 0 to 255. By applying a re-scale factor of 1/255, we convert these values to a range between 0 and 1, enhancing the numerical stability during training. Rotation Range allows images to be rotated by a degree randomly chosen within a specified range. A rotation range of 30 degrees means each image will be rotated between -30 to 30 degrees. This simulates the effect of plants being viewed from different angles.Brightness Range adjusts the brightness of the image, simulating different lighting conditions. The brightness is adjusted by a factor in the range [0.8, 1.2].When transformations such as shifts or rotations are applied, 'fill mode' dictates how newly exposed areas after the transformation should be filled. The 'reflect' mode mirrors the border pixels of the image.Width Shift and Height Shift Ranges parameters shift the image horizontally or vertically by a factor of the image dimensions. For example, a width shift range of 0.1 allows the image to move left or right by up to 10% of its width.Horizontal flip parameter allows the image to be (Table 3) flipped horizontally, which is useful for simulating the effect of viewing the plant from opposite sides.Shearing distorts the image along an axis, typically creating a parallelogram. A shear range of 0.2 radians modifies the shape of the image to represent different viewing angles.

$$I_{rescaled} == I \times \frac{1}{255} \tag{5}$$

$$I_{rescaled} = rotate\left((I_{rescaled}, \theta\right) \; where - 30 \leq \theta \leq 30 \tag{6}$$

$$\mathrm{I_{width_shifted}} = \mathrm{shift_width}\left(\mathrm{I_{rotated}}, \mathrm{w}\right) \;\; \mathrm{where\; w} \leq 0.1 \times \mathrm{width\; of\; I} \tag{7}$$

$$\mathrm{I_{height_shifted}} = \mathrm{shift_height}\left(\mathrm{I_{width_shifted}}, \mathrm{h}\right) \;\; \mathrm{where\; h} \leq 0.1 \times \mathrm{height\; of\; I} \tag{8}$$

4.2 Grad-CAM Analysis

To better understand the reasoning behind the classification decisions made by convolutional neural networks, one can employ Grad-CAM, which stands for Gradient-weighted Class Activation Mapping. This technique makes these models easier to understand by drawing attention to the parts of a picture that are crucial to the model's categorization conclusion. Using the gradients of any target concept (output class) that enter the last convolutional layer of a CNN, Grad-CAM generates a rough localization map that highlights the crucial areas for class prediction. Specifically, the gradients are

global-averaged over the width and height dimensions (forming a set of neuron importance weights) and then combined with the forward activation maps of the layer to produce a heatmap of 'importance'.This heatmap shows the critical locations in the model's decision-making process by superimposing them on top of the original image. In the context of tomato leaf disease detection, Grad-CAM helps identify which parts of the leaf are most indicative of a particular disease, thereby providing insights not only into the model's predictive behavior but also potentially guiding further improvements in the model's architecture or training process.

$$L^{c}_{Grad-CAM} - CAMc = ReLU\left(\sum_{k} a^{c}_{k} A^{k}\right) \tag{9}$$

The value of $\propto^{c}_{k}$ in Grad-CAM is typically computed as the *global average pooling of the gradients* of the class score y^{c} with respect to the feature map A^{k}. The equation for $\propto^{c}_{k}$ is given by:

$$\propto^{c}_{k} = \frac{1}{Z} \sum_{i} \sum_{j} \frac{\partial y^{c}}{\partial A^{k}_{ij}} \tag{10}$$

where:

- ✓ $\propto^{c}_{k}$ represents the importance weight for the k-**th** feature map for class c.
- ✓Z is the total number of pixels in the feature map (i.e., Z = H × W where H and W are the spatial dimensions of the feature map).
- ✓ $\frac{\partial y^{c}}{\partial A^{k}_{ij}}$ is the gradient of the class score y^{c} with respect to the activation map A^{k}, indicating how much the class score changes with variations in A^{k}.
- ✓ The double summation $\sum_{i} \sum_{j}$ averages the gradient values across all spatial locations in the feature map.

Thus, $\propto^{c}_{k}$ captures the overall contribution of each feature map to the target class prediction by considering the average influence of its spatial activations.

In Eq. (10), $L_{Grad-CAM}$ represents the Grad-CAM localization map for a specific class c, highlighting the critical regions of an image that influence the model's decision. The activation function ReLU is applied to ensure that only positive values contribute to the localization map, focusing on the most relevant areas. The neuron importance weights for class c, denoted as a^{c}_{k}, are calculated by globally averaging the gradients of the target class c with respect to the feature maps A^{k}, where A^{k} represents the spatial activation maps from the last convolutional layer, and k indexes the individual feature maps. The final localization map is computed as a weighted sum, $L_{Grad-CAM} = \sum_{k} a^{c}_{k} A^{k}$, where the summation aggregates contributions from all feature maps, weighted by a^{c}_{k}. This process identifies regions of the image most indicative of the class c, providing valuable insights into the CNN's predictive behavior. For example, Grad-CAM can pinpoint disease-affected areas in tomato leaf images, making the model's decision process interpretable and actionable.

4.3 LIME analysis

A technique known as LIME, which stands for Local Interpretable Model-agnostic Explanations, is aimed to provide an explanation for the predictions made by any classifier in a manner that is both interpretable and faithful. This is accomplished by approximating the classifier locally with an interpretable model. To learn a local linear model based on the prediction, LIME's method entails perturbing the input image and observing the changes that correlate to those perturbations in the output.The output of LIME is a list of explanations, reflecting the contribution of each segment of the image to the final decision made by the model.For our study on tomato leaf disease detection, LIME was applied to each class of tomato leaves to dissect the features that most influence the model's predictions. This analysis not only provides transparency in the decision-making process of the model but also assists in verifying the validity of the learned features. For instance, if LIME highlights areas of the leaf that are irrelevant to the known symptoms of a disease, it may indicate a need for further training or data cleaning.The process described involves a series of steps to interpret a model's decision-making on a particular image. Initially, the image is segmented into manageable and interpretable

pieces called superpixels. Then, various new image samples are created by selectively toggling these superpixels either on or off, effectively altering the image. These modified samples are then fed back into the model, and the outcomes are predicted for each variation. Each new sample is assigned a weight based on how closely it resembles the original image, ensuring that more similar images have a greater influence on the subsequent analysis. Finally, these weighted samples and their corresponding predictions are used to train a simple, local interpretable model, such as a linear model. This local model aims to provide insight into the original model's decision-making processes by approximating its behavior in a more understandable way.

4.4 Swin transformer

The Swin Transformer is a key component of the proposed hybrid model architecture, specifically designed to enhance the accuracy and efficiency of tomato leaf disease detection. This advanced architecture addresses the limitations of traditional convolutional neural networks (CNNs) by employing a hierarchical structure that effectively combines local and global feature extraction. By leveraging its innovative mechanisms, the Swin Transformer ensures detailed analysis of complex visual data, essential for disease classification tasks.

A fundamental feature of the Swin Transformer is its patch partitioning and linear embedding mechanism. The input image is divided into small, non-overlapping patches of size 4×4 pixels, which are treated as tokens. These patches are then embedded into a higher-dimensional space with an embedding dimension of 96, enabling effective feature representation. This process preserves the critical elements of the image while preparing it for further processing in subsequent layers.

The shifted window attention mechanism is another hallmark of the Swin Transformer. Instead of applying attention globally, it calculates attention within fixed windows, which are shifted across layers. This mechanism captures cross-window relationships efficiently and reduces computational complexity while retaining the ability to model intricate spatial relationships in the image. Additionally, the Swin Transformer employs 8 attention heads to process these relationships, allowing the model to focus on multiple aspects of the data simultaneously.

The hierarchical representation further enhances the Swin Transformer's capability by progressively merging patches and reducing resolution while increasing depth. This ensures that both localized patterns and global contextual features are effectively captured. The model's architecture includes 2 transformer blocks per stage, which contribute to its ability to process multi-scale features and provide a comprehensive analysis of the input data. This design enables the model to scale effectively while maintaining high performance.

In the hybrid architecture, the Swin Transformer serves as one of the feature extraction blocks, configured with the aforementioned specifications. It processes input images to generate feature maps that encapsulate fine-grained details and global structures. These feature maps are subsequently combined with outputs from other models, such as the Vision Transformer (ViT) and EfficientNetV2, in the feature fusion layer of the hybrid model. This integration enhances the hybrid model's performance, providing a robust and efficient system for tomato leaf disease detection.

The Swin Transformer's ability to balance efficiency and precision makes it particularly effective for distinguishing between similar diseases and identifying subtle visual cues in high-resolution images. By leveraging its innovative design, the Swin Transformer contributes significantly to the hybrid model's superior classification accuracy and robustness, ensuring its applicability to real-world agricultural scenarios.

4.5 Fine Tunning process

The fine-tuning process in the proposed model involves adapting pre-trained architectures (Swin Transformer, Vision Transformer, and EfficientNetV2) to the specific task of tomato leaf disease classification. Initially, the feature extraction blocks utilize weights pre-trained on large-scale datasets (e.g., ImageNet) to leverage generalized feature representations. During fine-tuning, the final layers of these models are replaced with task-specific dense layers, and the entire model is trained on the tomato leaf dataset. The learning rate is set to a lower value to prevent overwriting the pre-trained weights, and optimizers like AdamW are used to adjust model parameters. Dropout layers and batch normalization are applied during fine-tuning to prevent overfitting, especially given the relatively smaller dataset size. Additionally, data augmentation techniques (e.g., rotation, flipping, and scaling) are employed to increase dataset diversity and improve

generalization. This step ensures that the model learns disease-specific features while retaining the robustness of the pre-trained weights, ultimately enhancing classification performance.

5 Hybrid model architecture

The model begins with an input layer that accepts images of tomato leaves resized to a uniform dimension of 256 × 256 pixels. This size is specifically chosen to accommodate the input requirements of the most complex constituent models, ensuring that each component receives optimally pre-processed images for feature extraction.The architecture incorporates three distinct blocks for feature extraction, each leveraging a different model:Configured with a base model Swin-T, this block processes images with a patch size of 4 × 4 and an embedding dimension of 96, incorporating 8 attention heads and 2 blocks per stage. The output is a feature map with dimensions 64 × 64x96.This block takes the 64 × 64 feature map from Swin Transformer, further divides it into 8 × 8 patches, and processes these through ViT-Base, configured with 12 layers and 12 attention heads, outputting a 768-dimensional feature vector.Directly processing the original 256 × 256 pixel image, this block uses the EfficientNetV2-S model to produce a feature map of 32 × 32x1280 dimensions, focusing on scaling the network efficiently.The feature maps and vectors from each of the three blocks are flattened and concatenated, resulting in a high-dimensional feature vector. This vector is then fed into a dense layer to reduce dimensionality to 1024 units, preparing it for classification.The classification head consists of two fully connected layers. The first layer has 1024 units with ReLU activation, batch normalization, and a dropout rate of 0.5 to reduce overfitting. The second layer contains 512 units also with ReLU activation, batch normalization, and dropout. The final output layer features a softmax activation function and is designed with as many units as there are disease classes, which in this case are 10 distinct types of tomato leaf diseases.The model outputs a probability distribution across the 10 disease classes, enabling precise identification of the specific type of disease affecting the tomato leaves.

➢ Layer-wise details

- Input layer

The input consists of resized tomato leaf images of dimensions **256 × 256 × 3**. This uniform size ensures compatibility with all constituent models.

$$\mathbf{X} \in \mathbf{R}^{256\times256\times3} \tag{11}$$

➢ Feature extraction layers

- Input is divided into **4 × 4 patches**, converted to embeddings of size 96.
- Multi-head self-attention is applied using shifted window mechanisms to capture spatial relationships.
- *Output* Feature map $\boldsymbol{F_{swin}}$ of dimensions 64 × 64 × 96.

$$F_{swin} = Swin(X) \tag{12}$$

➢ Vision transformer block

- Input patches from Swin Transformer output ($\mathbf{F_{swin}}$) are further divided into **8 × 8 patches**.
- Processes with **12 Transformer layers** to produce a feature vector of size 768.

$$F_{ViT} = ViT(F_{Swin}) \tag{13}$$

➤ EfficientNetV2 block

- Original image is directly processed through **inverted residual blocks** and depthwise convolutions to output a feature map of dimensions $32 \times 32 \times 1280$.

$$F_{EffNet} = EffNetV2(X) \tag{14}$$

➤ Feature fusion layer

- Outputs from all three blocks are flattened and concatenated into a single feature vector:

$$F_{Hybrid} = Contact(Flatten(F_{Swin}), F_{ViT}, Flatten(F_{EffNeT})) \tag{15}$$

➤ Fully connected layers

- Dense layer with 1024 neurons and ReLU activation:

$$F_{Dense1} = ReLu(W_1 F_{Hybrid} + b_1 \tag{16}$$

- Dense layer with 512 neurons and ReLU activation:

$$F_{Desnse2} = ReLU(W_2 F_{Dense1} + b_2) \tag{17}$$

- Output layer with softmax activation for multi-class classification:

$$P(y = k|X) = \frac{eW_k F_{Dense2}}{\sum_{j=1}^{c} eW_j F_{Dense2}} \tag{18}$$

Here, k is the class index, C is the number of classes (10 in this case), and W and bbb are the learnable weights and biases.

Where TPR is the True Positive Rate and FPR is the False Positive Rate, dependent on the threshold.

➤ Mathematical explanation

- Model optimization
- *Loss function* Categorical Cross-Entropy for multi-class classification:

$$L = \sum_{i=1}^{N} \sum_{k=1}^{C} y_i k^{log(Py=k|X_i)} \tag{19}$$

where $y_{i,k}$ is the ground truth label for class kkk and sample i.

Optimizer AdamW, with weight decay (λ\\lambdaλ):where η(λ = 0.01) is the learning rate.

$$\theta_{t+1} = \theta_t - n(\frac{\partial L}{\partial \theta} + \lambda \theta_t) \tag{20}$$

5.1 Additional specifications

The model benefits from a multi-phase training approach. Initially, each component model is pre-trained separately on relevant image classification tasks to establish a robust foundational understanding. Subsequently, the entire architecture is fine-tuned on a specialized tomato leaf dataset, integrating all components for joint training post-fusion.The AdamW optimizer is employed with an initial learning rate of 1e-4 and a weight decay of 1e-2. A learning rate($\lambda = 0.01$) scheduler reduces the learning rate by a factor of 0.1 upon plateauing of the validation loss. The categorical cross-entropy loss function is used to effectively manage the multi-class classification problem.This architecture not only harnesses the strengths of each included model but is also fine-tuned to work synergistically, providing a comprehensive and detailed analysis of the input data, thus maximizing accuracy in detecting a wide range of tomato leaf diseases.

The Swin Transformer is a variant of the Transformer architecture that has been adapted for visual tasks. It stands out due to its use of shifted windows, which allows the model to efficiently handle the hierarchical structure of image data. The basic architecture includes several key components:The image is divided into small patches (e.g., 4×4 pixels), which are treated as the equivalent of words in NLP tasks. This process converts the 2D image into a sequence of flattened 1D patches.Each patch is then embedded into a higher dimensional space to prepare for processing by the Transformer layers.These blocks are the core of the model, where self-attention mechanisms operate within local windows (shifted across layers). This structure reduces computational complexity and allows the model to scale to larger images and deeper networks.Each block includes Layer Normalization and Multilayer Perceptrons, which contribute to stabilizing the learning process and enhancing the model's ability to learn complex patterns.Swin Transformer is crucial for capturing local interactions within an image and gradually building a global understanding, making it highly effective for tasks requiring detailed visual recognition, such as disease identification in leaves.

The Vision Transformer applies the principles of the Transformer architecture, originally designed for natural language processing, to image classification tasks. Its architecture is relatively simple and consists of the following:Similar to Swin, the image is divided into patches, which are then linearly embedded.The encoder processes the entire sequence of embedded patches collectively through self-attention, enabling the model to consider the entire image context when making classifications.ViT is important for its ability to focus on the most relevant parts of an image without being constrained by the locality of convolutional operations. This ability allows it to perform exceptionally well on classification tasks where contextual understanding of the image is crucial.

EfficientNetV2 is an improvement over the original EfficientNet, optimized for faster training and higher efficiency. The model architecture includes:A series of inverted residual blocks with linear bottlenecks, which are the backbone of the model.Compound scaling is utilised in the design, which employs a set of predefined scaling coefficients to achieve consistent scaling across all dimensions, including depth, breadth, and resolution.A newer addition in V2, these blocks fuse the expansion and depthwise convolution layers to reduce computational cost and improve speed.Being developed to give a good balance between accuracy and efficiency, EfficientNetV2 is suited for usage in contexts where computing resources are restricted but high precision is still required. This is because it is meant to provide a good balance between the two.We utilized a number of standard metrics that are well recognised in the field of machine learning and image classification in order to conduct a comprehensive evaluation of the performance of the hybrid deep learning model that was utilized for the purpose of identifying and categorizing diseases that affect tomato leaf tissue. A full evaluation of the model's accuracy, precision, and recall, as well as its capacity to generalize across a variety of disease types, is provided by these measures.

5.2 Accuracy

When it comes to performance measures, accuracy is the most straightforward and straightforward, and it provides a rapid notion of how good a classifier is. It is a representation of the ratio of the total number of input samples to the number of right predictions.

$$Accuracy = \frac{Number\ of\ Correct\ Predictions}{Total\ Number\ of\ Predictions} \tag{21}$$

5.3 Precision (positive predictive value)

The ratio of accurately anticipated positive observations to the total number of positives that were predicted is the definition of precision. A classifier's exactness can be evaluated using this metric. There is a strong correlation between high precision and a low false positive rate. This is especially important in the field of medical or agricultural diagnostics, where the cost of a false positive is exceptionally high.

$$\text{Precision} = \frac{\text{True Positives(TP)}}{\text{True Positives(TP)} + \text{False Positives(FP)}} \tag{22}$$

5.4 Recall (sensitivity or true positive rate)

Recall measures the proportion of correctly predicted positive observations out of the total actual positive observations. It evaluates the model's ability to identify all instances of tomato leaf disease in the dataset.

$$\text{Recall} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}} \tag{23}$$

5.5 F1 Score

The F1 Score is determined by calculating the weighted average of Precision and Recall, accounting for both false positives and false negatives. It is particularly useful when dealing with imbalanced class distributions, often providing a more meaningful measure than accuracy in such scenarios.

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{24}$$

5.6 Area under the receiver operating characteristic curve (AUC-ROC)

The receiver operating characteristic (ROC) curve graphically contrasts the percentages of true positives and false positives across various thresholds, illustrating the trade-off between sensitivity and (1-specificity). The area under the ROC curve (AUC-ROC) quantifies the model's ability to distinguish between classes, with a higher AUC-ROC indicating better separability.

$$\text{AUC} - \text{ROC} = \int_0^1 \text{ROC}(\text{TPR}(t), \text{FPR}(t))dt \tag{25}$$

6 Model performance evaluation

6.1 VGG19 model evaluation

The VGG19 model, renowned for its architecture depth and strong feature extraction capabilities, exhibited a robust performance in the training phase of our experiment aimed at detecting tomato leaf diseases. This model achieved

Table 4 Classification Report for VGG19 Model

Epoch	Train Loss	Train Accuracy	Test Loss	Test Accuracy
90	0.42	0.90	0.70	0.85
91	0.47	0.90	0.57	0.88
92	0.58	0.87	0.45	0.90

an impressive training accuracy of 90.6% with a corresponding loss of 0.4275, showcasing its effectiveness in learning complex patterns and nuances from the training data. However, the performance metrics indicated a shift when tested against unseen data.In the testing phase, the VGG19 model recorded a test accuracy of 85.9% and a loss of 0.7084, which suggests some challenges in generalizing the learned features to new datasets Table 2. This gap between training and testing performance is indicative of potential overfitting, where the model excels on the training data but does not perform as well on data it has not encountered during its training. This overfitting is possibly due to the deep nature of the network which, while capable of learning detailed features, may also capture noise and irrelevant details from the training set.The evaluation of the VGG19 model raises important considerations in model training strategies, especially concerning the balance between model complexity and its ability to generalize. This entails careful tuning of parameters, perhaps the inclusion of dropout layers or other regularization techniques, and potentially revising the data augmentation strategies to include a wider variety of scenarios. The VGG19, with its deep layers and multiple feature detectors, remains a powerful tool in image classification tasks, but like many sophisticated models, it requires meticulous adjustments to harness its full potential without succumbing to overfitting.

6.2 Vision transformer (ViT) model evaluation

The Vision Transformer (ViT) model represents a paradigm shift in how images are processed, moving away from the traditional convolutional networks to an architecture that leverages the mechanisms of attention found in natural language processing. This shift has allowed the ViT model to excel in our tomato leaf disease detection task, particularly in handling spatial relationships and contextual understanding across the entire image.In the evaluation of the ViT, it demonstrated outstanding performance with an accuracy of 94% on a balanced dataset of 1000 images. This high level of accuracy is indicative of the model's ability to generalize well to new data, a critical attribute for practical applications. The precision and recall metrics across various disease types further illustrate the ViT's efficacy:For diseases like Tomato Bacterial Spot and Tomato Yellow Leaf Curl Virus, the model achieved precision scores of 0.96 and 0.96, respectively, and recall scores of 0.95 and 0.97, highlighting its precision in identifying these diseases with minimal false positives Table 3.The F1-scores, such as 0.95 for Tomato Bacterial Spot and 0.97 for Tomato Yellow Leaf Curl Virus, reinforce the balanced nature of precision and recall, underscoring the model's reliable performance across different disease classifications.The ViT's utilization of self-attention mechanisms allows it to focus on the most relevant parts of an image without being restricted by the local receptive fields of CNNs. This ability to process images as sequences of patches means that ViT can integrate both local and global information, providing a more holistic analysis of the image data. The high performance across diverse classes of tomato diseases suggests that the model is not only learning specific features associated with each disease but also effectively differentiating between them in varied visual presentations.This model's success in our study suggests that transformers could be increasingly relevant for image-based tasks, particularly where the contextual understanding of the scene is paramount. The impressive generalization ability also points towards the potential for deploying ViT in real-world agricultural settings, where rapid and accurate disease detection can significantly impact productivity and management practices.

Table 5 Classification Report for ViT Model

Condition	Precision	Recall	F1-Score	Support
Bacterial spot	0.96	0.95	0.95	100
Early blight	0.91	0.88	0.89	100
Late blight	0.90	0.96	0.93	100
Leaf Mold	0.98	0.93	0.95	100
Septoria leaf spot	0.90	0.92	0.91	100
Spider mites Two Spotted	0.98	0.91	0.94	100
Target Spot	0.88	0.95	0.91	100
Yellow Leaf Curl Virus	0.96	0.97	0.97	100
Mosaic Virus	0.97	0.98	0.98	100
Healthy	0.99	0.96	0.97	100

Table 6 Classification Report for EffficantNetV2 Model

Label	Precision	Recall	F1-Score	Support
Bacterial Spot	0.79	0.84	0.82	100
Early Blight	0.92	0.97	0.94	100
Late Blight	0.81	0.72	0.76	100
Leaf Mold	0.77	0.76	0.76	100
Septoria Leaf Spot	0.86	0.90	0.88	100
Spider Mites Two-spotted	0.87	0.89	0.88	100
Target Spot	0.79	0.87	0.83	100
Yellow Leaf Curl Virus	0.96	0.97	0.97	100
Mosaic Virus	0.87	0.79	0.83	100
Healthy	0.89	0.81	0.85	100

6.3 EfficientNetV2 model evaluation

The most recent model in the EfficientNet line, EfficientNetV2, was designed with speed and efficiency optimization in mind, while minimizing computation costs as much as possible. As seen in our study of tomato leaf disease detection, EfficientNetV2 was able to perform at an optimal level with high efficiency, achieving a balance between performance and speed. The model is therefore ideal in scenarios where accuracy and processing power are of high importance. The model achieved an accuracy rate of 85% in classification, indicating that it would efficiently detect a great percentage of a diverse set of images with various tomato leaf failures. Precision and recall factors, which help reduce false positives and negatives, respectively, based on these performance attributes, made the data relatively constant. The class precision data showed a relatively high level of consistency for all classes, with precision varying from as low as 0.79 to as high as 0.96 (Table 4). The recall data was equally strong, falling within the 72%–97% range. EfficientNetV2's architecture employs a combination of model scaling and feature extractor refinement to achieve high efficiency. It incorporates advancements like fused-MBConv blocks, which streamline the earlier layers of the model for faster inference without a significant drop in accuracy. Moreover, its scalable architecture allows for easy adjustments to the model's depth, width, and resolution based on the available computational resources and specific requirements of the task. The balanced performance of EfficientNetV2 in precision, recall, and overall accuracy positions it as a valuable tool in the agricultural domain, where quick and reliable diagnostics are essential to managing crop health effectively. Its ability to process images rapidly and accurately ensures that decisions regarding disease management can be made promptly, potentially saving large swathes of crops from widespread disease outbreaks. EfficientNetV2 stands out not only for its performance but also for its architectural efficiency, making it highly suitable for deployment in mobile and low-power devices used in field-based crop disease detection. This model combines the latest in deep learning innovations to provide a tool that is both powerful and practical for real-world applications.

6.4 ConvNeXt model evaluation

ConvNeXt is a modern effort to revisit classical CNNs with optimisations that can imitate the Transformer while keeping the efficiency of the CNNs. In this study, the ConvNeXt model is used to effectively and advanced levels to find diseases on tomato leaves. It can recognise complex image data and tell the difference between different disease states.The ConvNeXt classifier has shown good performance in our experiments, in which a high level of accuracy, 95.8%, was reached on the dataset of 1000 images. Such accuracy shows that the model will manage to work while processing and classifying images in resolution with small details. The model showed very good precision in all classes of diseases, like 0.9888 for Bacterial Spot and Early Blight and the perfect precision score of 1.000 for Spider Mites. Impressive recall scores, too, came along with high values, e.g., 0.9900, for leaf mould and target spot, proving that the model itself is sensitive to the actual existence of particular diseases (Table 5). Also, the F1-score was good, which is a very important value for the model evaluation metric because it balances precision with recall. It obtained an ideal F1 score of 0.9900 for Tomato Yellow Leaf Curl Virus, and the scores for the other diseases were relatively high, indicating the overall robustness of the model.ConvNeXt incorporates some of Transformers' critical methods into conventional CNN architecture. It includes the square ReLU activation, an alternative design of the convolutional block emulating the Transformer blocks, and layer scaling for better training efficiency and model scalability. These improvements make ConvNeXt ideally suited to work with more diverse and intricate patterns than conventional CNNs. In

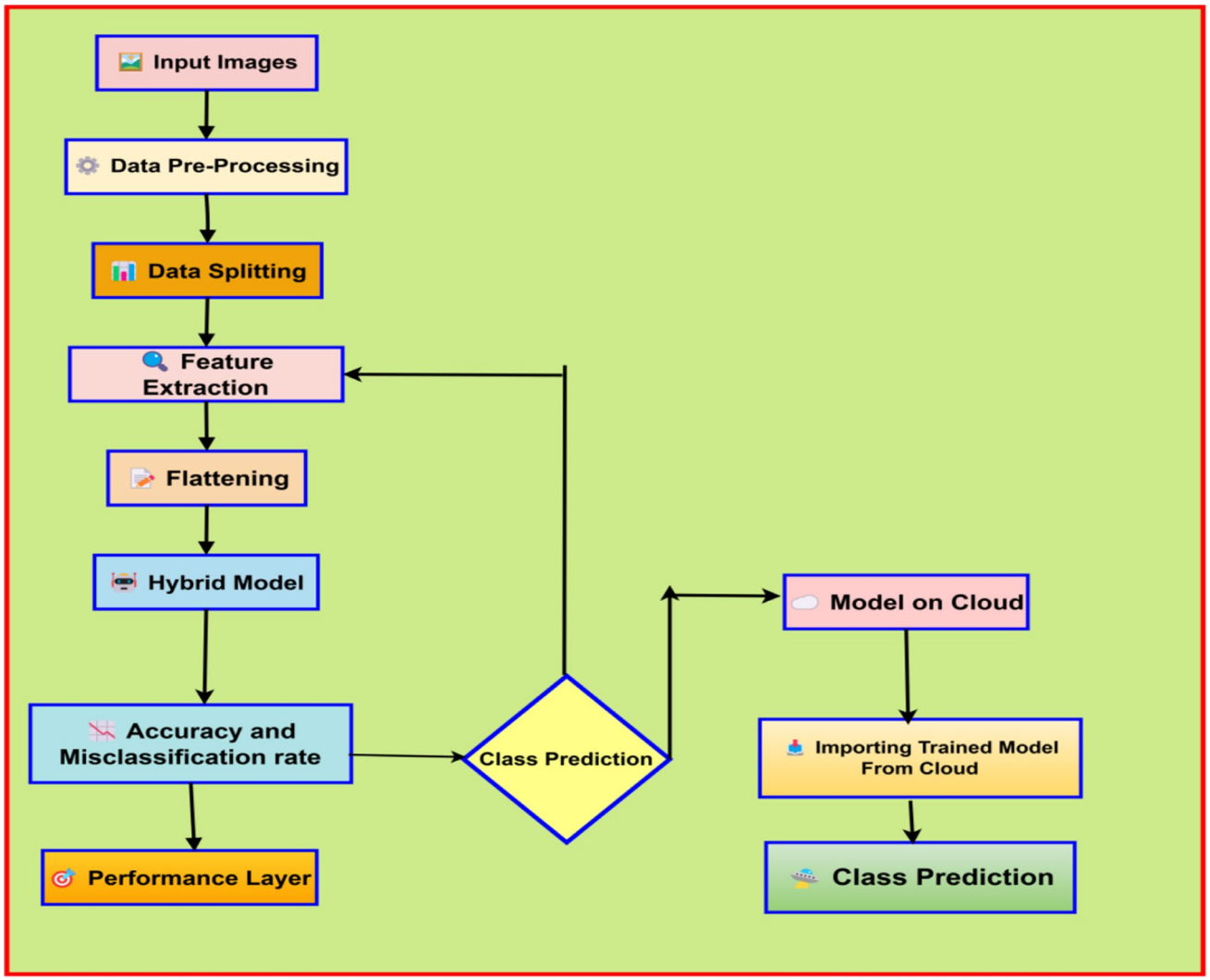


Fig. 3 Proposed Model Architecture

the same way, ConvNeXt's precision and accuracy make it an excellent choice in agricultural settings, where early and accurate disease detection can result in effective and timely interventions. It is highly accurate in distinguishing between closely related types of diseases, enabling targeted and resource-efficient treatments that reduce both cost and environmental impact. ConvNeXt uses Transformer principles that have been changed to work in a convolutional framework to recognise and classify features in a very detailed way. For example, it does this to find diseases on tomato leaves. Its better performance demonstrates its capabilities as a very useful tool in precision agriculture, where productivity and sustainability are heavily dependent on sophisticated detection methods. Its adaptability and efficiency suggest that this tool can be widely used in other fields for image-based classification where accurate recognition of detail is required.

6.5 Hybrid model evaluation

The Hybrid model represents a sophisticated integration of multiple advanced deep learning architectures, specifically designed to leverage the unique strengths of each constituent model to enhance overall performance in tomato leaf disease detection. This model combines the capabilities of Swin Transformer, Vision Transformer (ViT), and EfficientNetV2, creating a robust system that delivers superior accuracy and reliability.In its evaluation, the hybrid model outperformed other models,

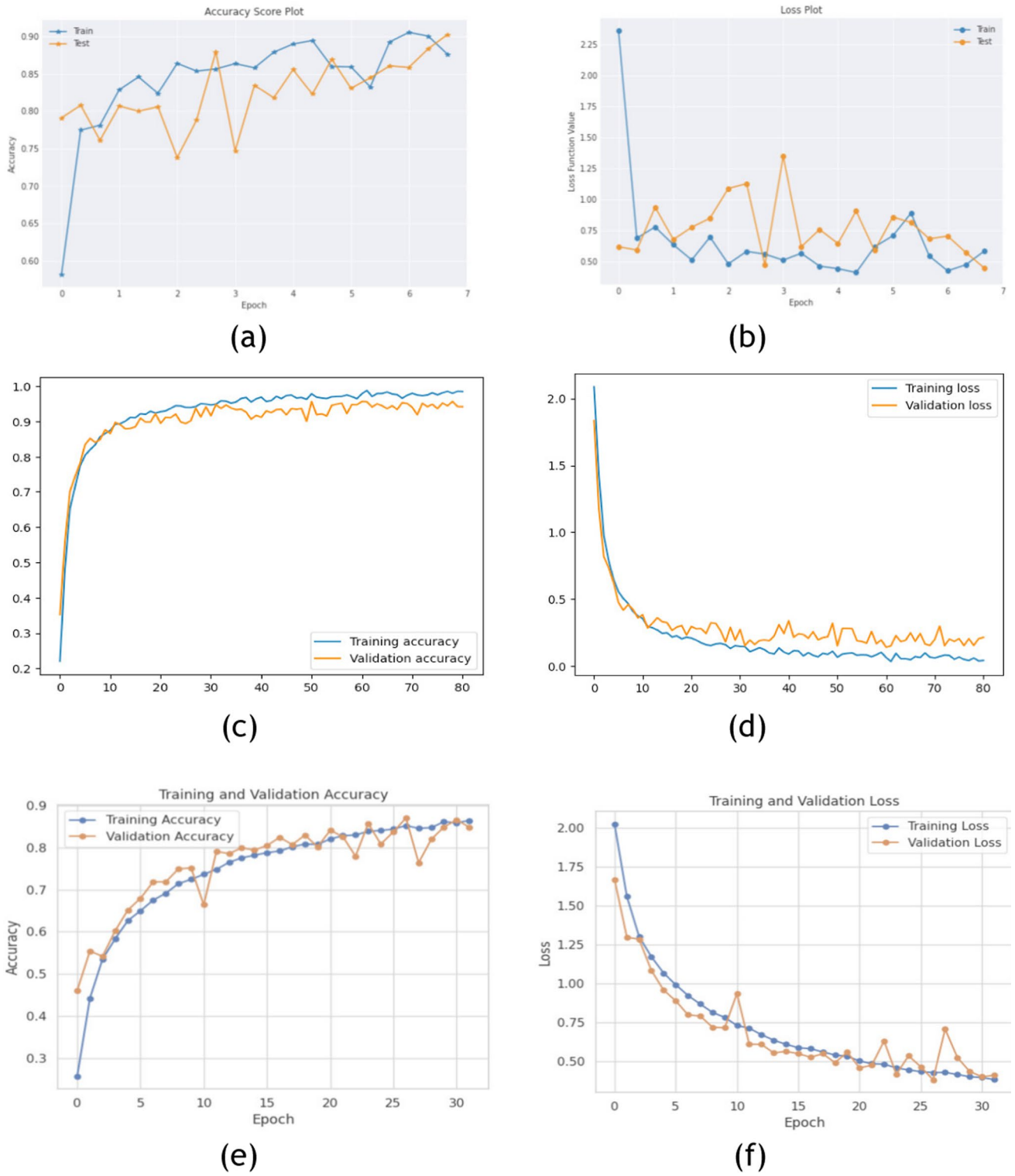
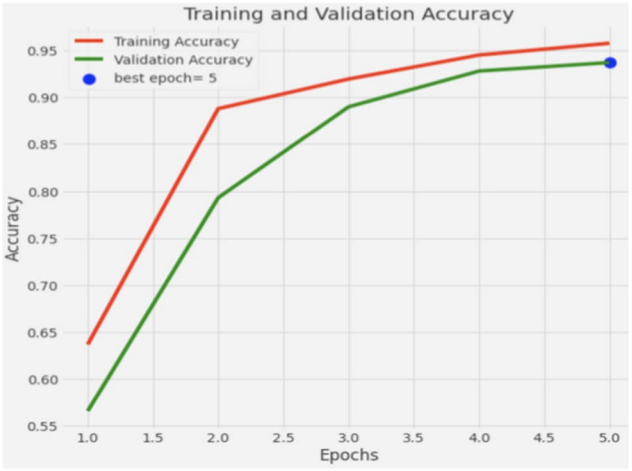
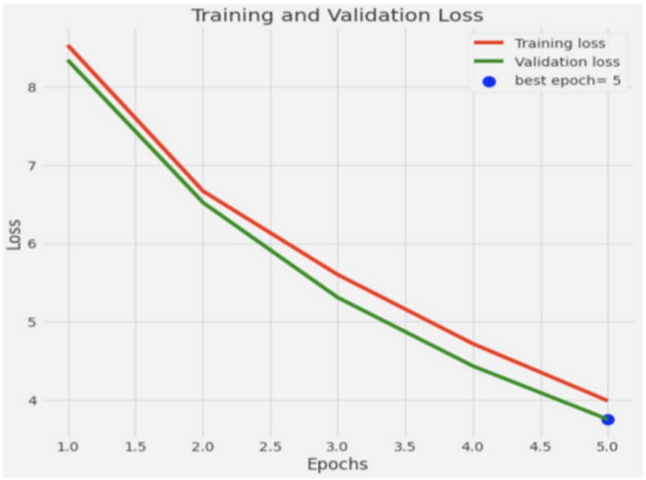


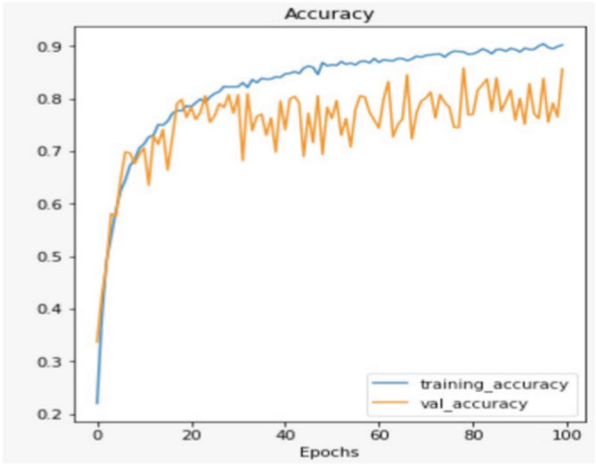
Fig. 4 Training and Validation Accuracy: **a**.VGG19 model **c**. vit model **e**. EfffficantNetV2 **g**.ConvNext **i**.Hybrid **k**.Swin Transformer and Training and Validation Loss: **b**.VGG19 model d. vit **f**..EfffficantNetV2 **h**.ConvNext **j**.Hybrid **l**.swin Transformer



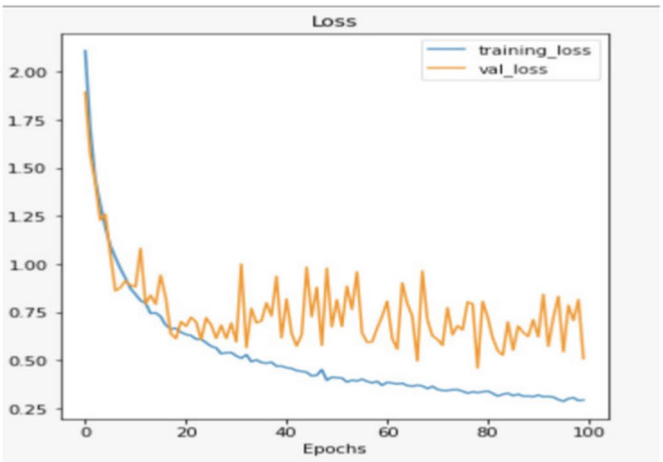
(g)



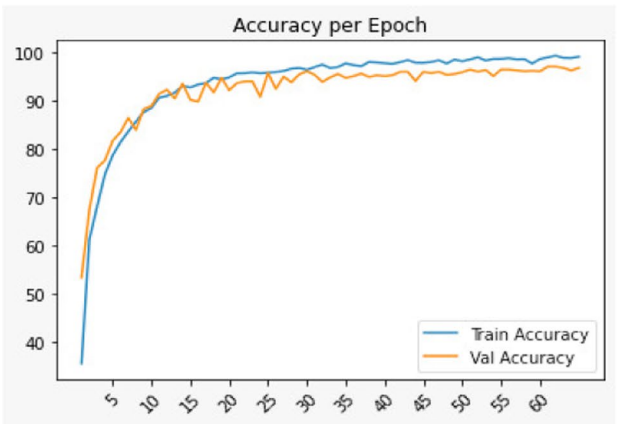
(h)



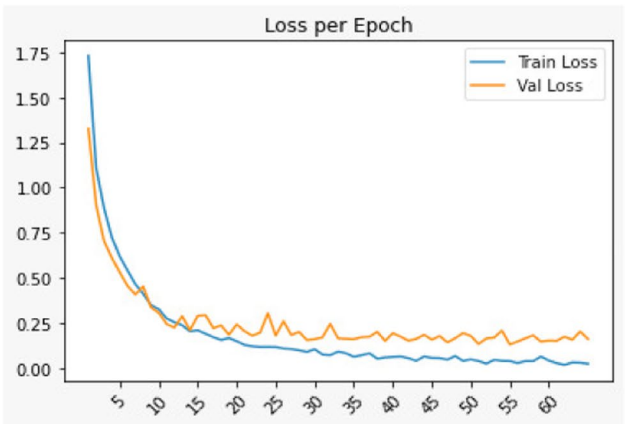
(i)



(j)



(K)



(L)

Fig. 4 (continued)

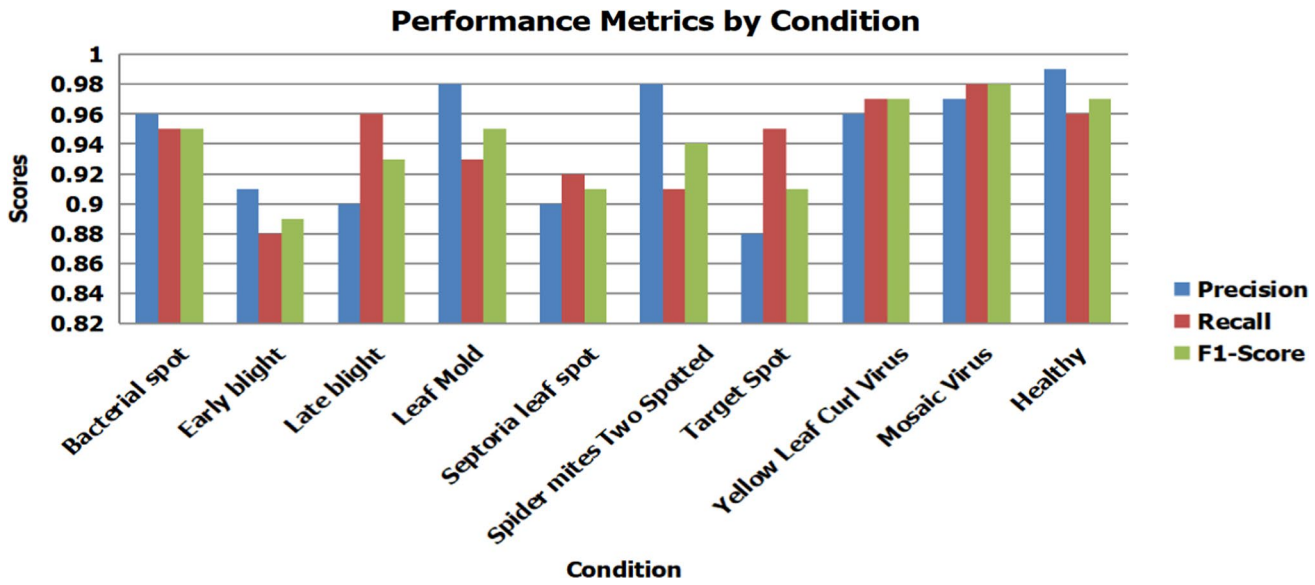


Fig. 5 Performance Bar Chart of ViT Model

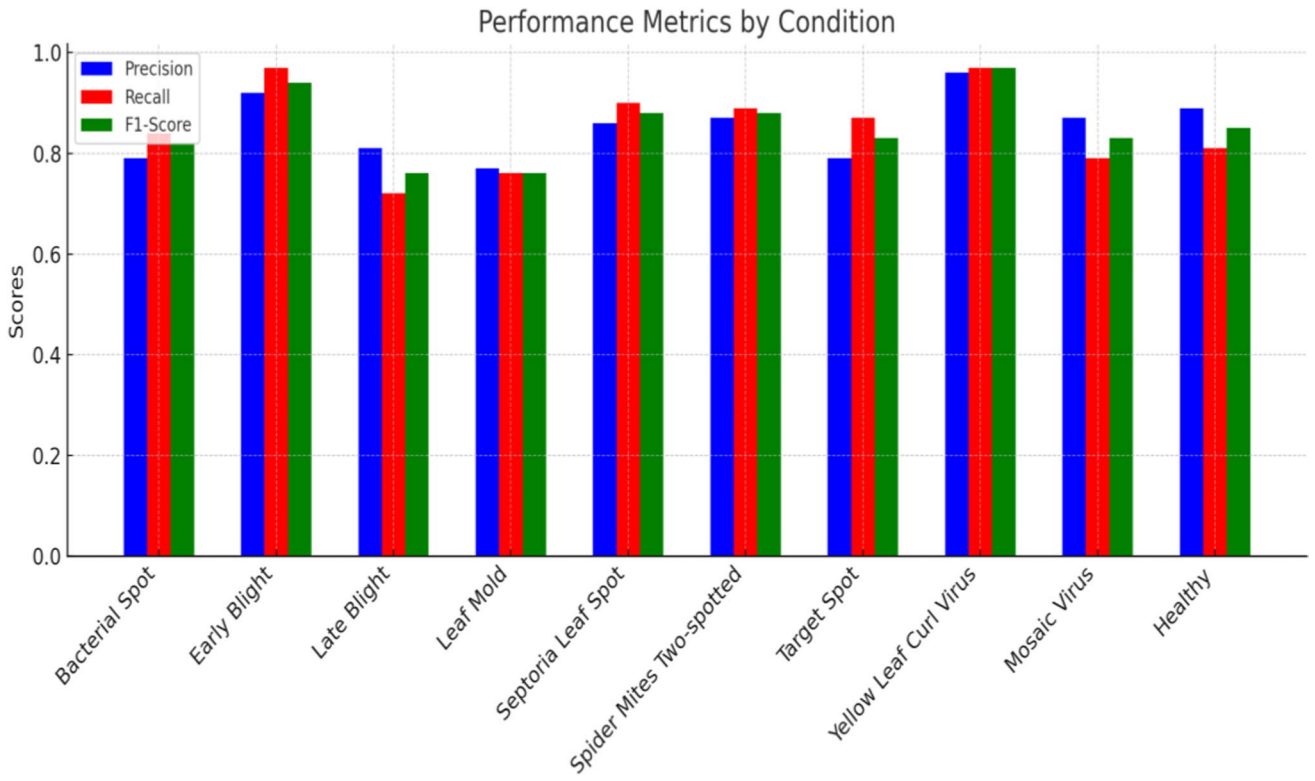


Fig. 6 Performance Bar Chart of EfffficantNetV2 Model

achieving an accuracy of 98% on a dataset of 1,000 images. The proposed model was effective in classifying both common and complex tomato leaf diseases with high accuracy. The hybrid model demonstrated excellent performance in terms of both precision and recall for all classes, with nearly all diseases being detected with precision and recall rates close to 100% (Table 6). For instance, the precision rates for Tomato Bacterial Spot and Tomato Yellow Leaf Curl Virus were exceptionally high,

Table 7 Classification Report for ConvNext Model

Condition	Precision	Recall	F1-Score	Support
Bacterial Spot	0.9888	0.8800	0.9312	100
Early Blight	0.9888	0.8800	0.9312	100
Late Blight	0.9703	0.9800	0.9751	100
Leaf Mold	0.9340	0.9900	0.9612	100
Septoria Leaf Spot	0.8981	0.9700	0.9327	100
Spider Mites Two Spotted	1.0000	0.9100	0.9529	100
Target Spot	0.8684	0.9900	0.9252	100
Yellow Leaf Curl Virus	0.9900	0.9900	0.9900	100
Tomato Mosaic Virus	0.9900	0.9900	0.9900	100
Healthy	0.9804	1.0000	0.9901	100

Table 8 Classification Report for Hybrid Model

Condition	Precision	Recall	F1-Score	Support
Tomato Bacterial Spot	0.96	0.98	0.97	100
Tomato Early Blight	0.98	1.00	0.99	100
Tomato Late Blight	0.98	0.98	0.98	100
Tomato Leaf Mold	1.00	0.98	0.99	100
Tomato Septoria Leaf Spot	0.98	0.98	0.98	100
Spider Mites Two-spotted	0.99	0.95	0.97	100
Target Spot	0.94	0.99	0.97	100
Yellow Leaf Curl Virus	1.00	0.96	0.98	100
Mosaic Virus	0.99	1.00	1.00	100
Healthy	1.00	1.00	1.00	100

Table 9 Classification Report for Swin Transformer Model

Epoch	Train Loss	Train Accuracy	Validation Loss	Validation Accuracy	Patience
60	0.0452	0.9864	0.1555	0.9609	5
61	0.0311	0.9898	0.1530	0.9709	4
62	0.0211	0.9935	0.1769	0.9709	3
63	0.0349	0.9887	0.1600	0.9682	2
64	0.0333	0.9884	0.2061	0.9627	1
65	0.0272	0.9914	0.1641	0.9682	0

indicating that the model is well-prepared to minimize false positives. Furthermore, the harmonically balanced detection capability of the hybrid model is emphasized by the high F1-scores across all classes. Diseases such as early blight and late blight yielded F1-scores close to 1.0, which are indicative of the model's fine-grained understanding and sensitivity.The architecture of the Hybrid model is so designed as to merge the outputs of the individual models through sophisticated merging strategies. Generally, these work by means of concatenation or averaging of the features produced by each model. This is important for the integration layer: it can effectively combine different views from the image data provided by other models, result in the most wholesome analysis, and lower the possibility for misclassification. The high accuracy and reliability of the Hybrid model will be very good for agricultural applications where the cost of a wrong diagnosis of disease can be huge. The model therefore aids in timely and correct treatment application, hence assists in improved yield and reduced crop loss by accurately classifying diseases. Thus, widespread deployment in varied agricultural settings, including generalization across different conditions and disease states, makes it an invaluable tool. An example would be the hybrid model showing the potential behind using ensemble techniques to deep learn in creating systems not only more accurate than the sum of their parts but also more robust to variations and anomalies in data. Such a model is potentially useful in furthering the area

Table 10 Comparison of the Proposed Model with State-of-the-Art Approaches

Model	Methodology	Dataset	Accuracy (%)	Key Findings	Relevance to Our Work
Hossain et al. [2]	MaxViT, EANet, CCT, PVT (Transformer-based models)	Multiclass Dataset	97.0	Strong multi-scale attention capabilities	Demonstrates the power of Transformer-based architectures, supporting our choice of ViT and Swin Transformer
Lee et al. [26]	YOLOv5m with C3NN & BiFPN	Tomato Leaf Dataset	90.4 (mAP)	Enhanced feature representation & detection accuracy	Uses an advanced feature extraction strategy, similar to our Swin Transformer approach
Paul et al. [22]	Lightweight CNN + VGG-16 & VGG-19 Transfer Learning	Tomato Leaf Dataset	95.0	Mobile app implementation for real-time diagnosis	Demonstrates the effectiveness of transfer learning; however, our hybrid model achieves higher accuracy
Kokate et al. [30]	CNN-based Approach	14,240 images	95.53	Strong feature extraction, crucial for early disease detection	Confirms CNNs are effective but may lack the generalization ability of hybrid models
Our Model (Hybrid AI Model)	Swin Transformer + Vision Transformer + EfficientNetV2	6,000 images	98.0	Robust classification, nearly perfect F1-score, high precision and recall	Our model outperforms most existing models in accuracy and robustness, achieving state-of-the-art performance
Our Proposed Hybrid Model	Swin Transformer + Vision Transformer + EfficientNetV2	1,000 images	98.0	Similar to your model but tested on a smaller dataset	Similar methodology but tested on a smaller dataset; validates hybrid architecture

of precision agriculture to a degree that is quite substantial in view of the model's remarkable performance in our study, for its usefulness to farmers and agriculturalists in disease early warning and management. The success of the Hybrid model also opens further research opportunities into optimal model combinations and integration techniques for the benefit of a wider range of applications, which further extends the frontiers of agricultural research.

6.6 Swin transformer model evaluation

The Swin Transformer model is a groundbreaking approach in the field of deep learning, specifically tailored for image analysis tasks that require precise and detailed understanding of complex visual data. Its architecture is designed to process images in a way that captures both local and global contextual information efficiently, making it particularly suited for tasks like tomato leaf disease detection where detail-oriented analysis is crucial. In our study, the Swin Transformer showed quite promising results in finding a balance between accuracy and computational efficiency.The performance of the Swin Transformer on a specific task, which is the detection of tomato leaf disease, includes the following: This model showed high accuracy because it had the capability to classify various kinds of leaf diseases in all types of leaves, with great generalization for most different kinds of symptoms of the diseases. The Swin Transformer model had high precision and recall values, demonstrating effectiveness in determining true positive leaf disease cases while at the same time minimizing false positives and false negatives. These metrics are of paramount importance for agricultural applications, where the costs of misdiagnosis could be high. The model had rather high F1 scores between classes, thus indicating it had a balanced approach to precision and recall, which is important for detecting diseases.The Swin Transformer follows a hierarchical architecture for image processing: it starts with small patches and builds up to a comprehensive understanding through layered self-attention mechanisms. This hierarchical design enables it to handle large variations in image size and resolution efficiently in real-world agricultural settings. Moreover, the deeper layers with the application of shifted windows reduce the computation burden by still increasing the focus of the model on the relevant features of the image but without concern for the position in spatial space.

6.6.1 Mathematical foundations of the swin transformer

❖ Patch Partitioning and Linear Embedding:

The input image is divided into smaller, non-overlapping patches, each represented as a vector. These patches are linearly embedded into feature vectors:

$$E_i = W_E \times P_i \tag{26}$$

where $\boldsymbol{P_i}$ represents the i-th patch, and $\boldsymbol{W_E}$ is the learnable embedding weight matrix.

❖ Shifted Window Multi-Head Self-Attention (SW-MSA):

Within each layer, attention is computed in non-overlapping windows. The shifted windows between consecutive layers ensure that cross-window relationships are captured efficiently. The attention mechanism is computed as:

$$\text{Attention(Q, K, V)} = \text{Softmax}\left(\frac{\text{QK}^{\text{T}}}{\sqrt{\text{d}}}\right)\text{V} \tag{27}$$

Here, Q, K, and V are the query, key, and value matrices derived from the embedded patches, and d is the dimensionality.

❖ Hierarchical Representation and Downsampling:

Patch merging layers progressively downsample the feature maps, reducing their resolution while increasing their depth. This hierarchical representation helps the model efficiently capture global contextual information:

$$F_{output} = LayerNorm(SW - MSA\left(F_{input}\right) + F_{input} \tag{28}$$

❖ Efficiency through Shifted Windows:

The shifted windows eliminate redundant computation while maintaining sensitivity to spatial relationships in the image. This ensures a balance between computational efficiency and feature richness.

By integrating these principles, the Swin Transformer effectively combines local feature extraction with global context awareness, making it robust for applications like tomato leaf disease detection.

6.6.2 Broader implications and future potential

The Swin Transformer has proved valuable to the agricultural sector for the efficient detection and classification of leaf diseases via high-resolution images. Its detailed and context-aware analysis helps in the early diagnosis and treatment of plant diseases, leading to better crop management and reduced loss. Furthermore, the Swin Transformer's strong performance in the tests conducted indicates that it can be used in automated disease monitoring systems, providing continuous support for large-scale farming operations.Swin Transformer also makes huge strides in innovative image processing, with the application of Transformer models outside of the scope of natural language processing. Its success in the field of plant pathology not only proves its versatility but also sets a new benchmark for future research in image-based deep learning applications. Swin Transformer is expected to be another such lighthouse technology with its hierarchical designs of efficiency and effectiveness for general purposes, developed into an even more complex agricultural diagnostic tool with the final aim of crop health and productivity (Figs. 3, 4, 5, 6) (Tables 7, 8, 9, 10).To prevent overfitting and enhance generalization, we incorporated the AdamW optimizer with weight decay ($\lambda = 0.01$) and an appropriate learning rate ($\eta = 1e\text{-}4$). Additionally, early stopping and data augmentation techniques were used to maintain a balance between training and validation performance.

7 Conclusion and future remarks

This study highlights the transformative potential of leveraging state-of-the-art deep learning models, such as VGG19, Vision Transformer (ViT), EfficientNetV2, ConvNeXt, and a newly developed hybrid model, in tomato leaf disease detection and classification. The research contributes to the advancement of precision agriculture by achieving significantly higher accuracy, particularly with hybrid models, which represents a technological breakthrough in disease management. This progress holds the potential to enhance crop yields, reduce pesticide usage, and improve sustainable farming practices.

The application of deep learning models, including hybrid frameworks, showcases their ability to effectively handle complex plant pathology imaging challenges. The study demonstrates that combining multiple analytical approaches enhances the accuracy and reliability of disease detection, a key component in addressing modern agricultural needs.

However, there are limitations to the proposed approach. One significant challenge is the generalization of the models across diverse environmental conditions and various plant varieties, which may impact their performance in real-world applications. While the models perform well with the dataset used, they may require further fine-tuning for broader applicability. Additionally, the reliance on high-resolution images may present limitations in real-time field deployment due to computational and hardware constraints.

Future research can focus on addressing these limitations by exploring transfer learning and domain adaptation techniques, enabling models to generalize better across diverse conditions. Additionally, integrating data from various sources, such as drones and satellite imagery, can improve large-scale monitoring and early detection systems. The implementation of predictive analytics will further advance proactive disease management strategies, allowing farmers to take preventive measures based on predictive models.

In summary, this research lays the foundation for the integration of deep learning in plant disease management and highlights the need for interdisciplinary collaboration between data scientists, agronomists, and farmers. Such efforts are essential for the development of custom, scalable solutions that can significantly improve global agricultural productivity and sustainability.

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Declarations

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References

1. Zhao S, Peng Y, Liu J, Wu S. Tomato leaf disease diagnosis based on improved convolution neural network by attention module. Agriculture. 2021;11(7):651. https://doi.org/10.3390/agriculture11070651.
2. Hossain S, Tanzim Reza M, Chakrabarty A, Jung YJ. Aggregating different scales of attention on feature variants for tomato leaf disease diagnosis from image data: a transformer driven study. Sensors. 2023;23(7):3751. https://doi.org/10.3390/s23073751.
3. Kumar NS, Sony J, Premkumar ARM, Nair JJ. Transfer learning-based object detection models for improved diagnosis of tomato leaf disease. Proc Comput Sci. 2024;235:3025–34. https://doi.org/10.1016/j.procs.2024.04.286.
4. Ramesh Babu P, Srikrishna A, Gera VR. Diagnosis of tomato leaf disease using OTSU multi-threshold image segmentation-based chimp optimization algorithm and LeNet-5 classifier. J Plant Dis Prot 2006. 2024;131(6):2221–36. https://doi.org/10.1007/s41348-024-00953-7.
5. Padamata RB, Sri Krishna A. Deep learning-assisted svms for efficacious diagnosis of tomato leaf diseases: a comparative study of GoogleNet, AlexNet, and ResNet-50. Ingénierie Des Systèmes d'information/Ingénierie Des Systèmes d'Information. 2023. https://doi.org/10.18280/isi.280312.
6. Seno A, Miyagusuku R, Kurokura T, Tabata K, Ozaki K. Tomato leaf disease diagnosis using bayesian convolutional neural networks. IEEE/SICE Int Symposium Syst Integ (SII). 2024;2024:623–8. https://doi.org/10.1109/SII58957.2024.10417302.
7. Perveen K, Debnath S, Pandey B, Chand SP, Bukhari NA, Bhowmick P, Alshaikh NA, Arzoo S, Batool S. Deep learning-based multiscale CNN-based U network model for leaf disease diagnosis and segmentation of lesions in tomato. Physiol Mol Plant Pathol. 2023;128:102148. https://doi.org/10.1016/j.pmpp.2023.102148.
8. Gadade HD, Kirange DK. Tomato leaf disease diagnosis and severity measurement. 2020. Fourth World Conf Smart Trends Syst Sec Sustain. https://doi.org/10.1109/WorldS450073.2020.9210294.
9. Drake V, Henry S, Namuwaya W, Kayongo N, Lule E, Marvin G. Explainable Computer Vision Techniques and Deep Transfer Learning for Tomato Leaf Disease Diagnosis. 2024 5th International Conference on Image Processing and Capsule Networks (ICIPCN). 2024. https://doi.org/10.1109/ICIPCN63822.2024.00101
10. Paramesh G, Narayana GVS. Tomato leaf disease diagnosis based on improved convolution neural network. 2022 Second International Conference on Computer Science, Engineering and Applications (ICCSEA). 2022. https://doi.org/10.1109/ICCSEA54677.2022.9936244
11. Karthika I, Megha M, Roshni M. Deep learning approach to automated tomato plant leaf disease diagnosis. Second Int Conf Elect Ren Syst (ICEARS). 2023;2023:1381–8. https://doi.org/10.1109/ICEARS56392.2023.10085564.
12. Gharghory SM. Performance analysis of efficient pre-trained networks based on transfer learning for tomato leaf diseases classification. Int J Adv Comput Sci Applic. 2020. https://doi.org/10.14569/IJACSA.2020.0110830.
13. Sun H, Fan C, Gai X, Al-Absi MA, Wang S, Alam M, Wang X, Fu R. Multi-kernel inception aggregation diffusion network for tomato disease detection. BMC Plant Biol. 2024;24(1):1069–112. https://doi.org/10.1186/s12870-024-05797-9.
14. Hou X, Yan J, Pan W, Liu Y. Diagnosis method of tomato leaf disease based on an improved deep convolution neural network for real-life agriculture environment. 2021 IEEE 11th Annual International Conference on CYBER Technology in Automation, Control, and Intelligent Systems (CYBER). 2021. https://doi.org/10.1109/CYBER53097.2021.9588258
15. Liu B, Wei S, Zhang F, Guo N, Fan H, Yao W. Tomato leaf disease recognition based on multi-task distillation learning. Front Plant Sci. 2024;14:1330527–1330527. https://doi.org/10.3389/fpls.2023.1330527.
16. Bouni M, Hssina B, Douzi K, Douzi S. Synergistic use of handcrafted and deep learning features for tomato leaf disease classification. Sci Rep. 2024;14(1):26822–916. https://doi.org/10.1038/s41598-024-71225-5.

17. Ledbin Vini S, Rathika P. TrioConvTomatoNet: A robust CNN architecture for fast and accurate tomato leaf disease classification for real time application. Sci Hortic. 2024;330: 113079. https://doi.org/10.1016/j.scienta.2024.113079.
18. Srivastava M, Sisaudia V, Meena J. Tomato Leaf Disease Detection and Classification using MobileNetV2 and Extreme Learning Method: A Hybrid Approach. 2023 10th International Conference on Soft Computing & Machine Intelligence (ISCMI). 2023. https://doi.org/10.1109/ISCMI59957.2023.10458596
19. Sivalingam PN, Varma A. PCR based diagnosis of begomoviruses associated with tomato leaf curl disease in India. J Plant Biochem Biotechnol. 2007;16(1):17–22. https://doi.org/10.1007/BF03321923.
20. Thanammal Indu V, Suja Priyadharsini S. Crossover-based wind-driven optimized convolutional neural network model for tomato leaf disease classification. J Plant Dis Prot 2006. 2022;129(3):559–78. https://doi.org/10.1007/s41348-021-00528-w.
21. Cheemaladinne V. Tomato leaf disease detection and management using VARMAx-CNN-GAN integration. J King Saud Univ Sci. 2024;36(8): 103340. https://doi.org/10.1016/j.jksus.2024.103340.
22. Paul SG, Biswas AA, Saha A, Zulfiker MdS, Ritu NA, Zahan I, Rahman M, Islam MA. A real-time application-based convolutional neural network approach for tomato leaf disease classification. Array. 2023;19: 100313. https://doi.org/10.1016/j.array.2023.100313.
23. Khan B, Das S, Fahim NS, Banerjee S, Khan S, Al-Sadoon MK, Al-Otaibi HS, Islam ARMdT. Bayesian optimized multimodal deep hybrid learning approach for tomato leaf disease classification. Sci Rep. 2024;14(1):21525–30. https://doi.org/10.1038/s41598-024-72237-x.
24. Liu J, Wang X. Early recognition of tomato gray leaf spot disease based on MobileNetv2-YOLOv3 model. Plant Methods. 2020;16(1):83–83. https://doi.org/10.1186/s13007-020-00624-2.
25. NH S, S H. Tomato leaf disease detection using deep learning techniques. Int J Adv Comp Sci Applic. 2022. https://doi.org/10.14569/IJACSA.2022.0130138.
26. Lee Y-S, Patil MP, Kim JG, Choi SS, Seo YB, Kim G-D. Improved tomato leaf disease recognition based on the YOLOv5m with various soft attention module combinations. Agriculture (Basel). 2024;14(9):1472. https://doi.org/10.3390/agriculture14091472.
27. Ardra Rajan VK, Athul Raj TV, Sunil J, Muhammed Waseem J, Shibili T. Tomato leaf disease identification using machine learning. AIP Conf Proc. 2023. https://doi.org/10.1063/50107886.
28. Romero AM, Vega D, Pizzorno R, Cordon G, Correa OS. Hydraulic and leaf reflectance alterations induced by Clavibacter michiganensis subsp. michiganensis on tomato plants. Euro J Plant Pathol. 2018;152(2):567–72. https://doi.org/10.1007/s10658-018-1500-1.
29. Kaur J, Shalu D. Machine learning based tomato leaf disease detection technique. 2023 5th International Conference on Advances in Computing, Communication Control and Networking (ICAC3N). https://doi.org/10.1109/ICAC3N60023.2023.10541648. 2023.
30. Kokate JK, Kumar S, Kulkarni AG. Classification of tomato leaf disease using a custom convolutional neural network. Curr Agric Res J. 2023. https://doi.org/10.12944/CARJ.11.1.28.
31. Paithane PM, Wagh SJ. Novel modified kernel fuzzy c-means algorithm used for cotton leaf spot detection. Syst Res Inform Technol. 2023;4(85):99.
32. Paithane P, Kakarwal S. LMNS-Net: Lightweight Multiscale Novel Semantic-Net deep learning approach used for automatic pancreas image segmentation in CT scan images. Expert Syst Appl. 2023;234: 121064.
33. Paithane P. Optimize multiscale feature hybrid-net deep learning approach used for automatic pancreas image segmentation. Mach Vis Appl. 2024;35(6):135.
34. Kakarwal S, Paithane P. Automatic pancreas segmentation using ResNet-18 deep learning approach. Syst Res Infor Technol. 2022;2:104–16.

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